RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105

DEPARTMENT OF INFORMATION TECHNOLOGY

CS23621 - MOBILE APPLICATION AND DEVELOPMENT LABORATORY
LABORATORY MANUAL

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Academic Year :	2025 - 26

RAJALAKSHMI ENGINEERING COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: ……GOKUL KRISHNA R……

Academic Year: …2025 - 26…. Semester: …VI…… Branch: B.TECH IT

Register No.	**231001046**

Certified that this is the Bonafide record of work done by the above student in the……CS23621 - MOBILE APPLICATION AND DEVELOPMENT…… Laboratory during the academic year 2026 – 2027.

Signature of Faculty in-charge

Submitted for the Practical Examination held on ……

Internal Examiner External Examiner

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EXPT NO.	EXPERIMENT NAME	GITHUB QR
1	Design a simple Android app using Kotlin basics to display a welcome message using variables, data types, and control statements.	
2	Create an Android app to perform arithmetic operations using functions for addition, subtraction, multiplication, and division.	
3	Develop an Android app to manage student details using classes and objects in Kotlin.	
4	Build your first Android app with a button and handle click events to display a toast message.	
5	Design a login screen using LinearLayout or ConstraintLayout with username and password fields.	
6	Create navigation between Login and Home screens using intents or navigation components.	
7	Demonstrate Activity and Fragment lifecycle methods and display lifecycle changes on the screen.	
8	Develop an Android app using MVVM architecture with ViewModel and display a list of items.	
9	Build an Android app with Room database for data persistence to store and retrieve data.	
10	Create an Android app using RecyclerView to display a list of items with images and click events.	
11	Develop an Android app to fetch data from an API and display the fetched data.	
12	Implement Repository Pattern and WorkManager for efficient background data fetching and caching.	
13	Design a user-friendly Android app UI with proper colors, spacing, alignment, and usability principles.	

Expt No : 1 **Date:**

Kotlin Basics

Aim

Tostudy and implement basic Kotlin programming concepts such as variables, data types, operators, control statements, and null safety.

Algorithm

1. Startthe Android Studio IDE.
2. Create a new Kotlin project.
3. Declare variables using val and var.
4. Use basic data types such as Int, Double, String, and Boolean.
5. Implement conditional statements (if, when).
6. Implement loops (for, while).
7. Display output using println() or UI components.
8. Run the program and verify output.

Coding

```
package com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview
import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!"
fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) {
```

```
 fun introduce(): String = "Hi, I'm $name and I'm $age years old"
}

data class User(val id: Int, val username: String, val email: String)

enum class Priority(val value: Int) {
    LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
    fun getDisplayName(): String = name.lowercase().replaceFirstChar { it.uppercase() }
}

object AppConfig {
    const val APP_NAME = "Kotlin Demo"
    const val VERSION = "1.0.0"
    fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
    companion object {
        const val PI = 3.14159
        fun circleArea(radius: Double): Double = PI * radius * radius
    }
}

abstract class Animal(val name: String) {
    abstract fun makeSound(): String
    fun sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) {
    override fun makeSound(): String = "$name says: Woof!"
}

interface Drawable {
    fun draw(): String
}

class Circle(private val radius: Double) : Drawable {
    override fun draw(): String = "Drawing a circle with radius $radius"
}

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
```

```
                    modifier = Modifier.padding(innerPadding)
                )
            }
        }
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column(
        modifier = modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Kotlin Fundamentals Demo",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") {
            demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
```

```
        }

        // Classes and Objects
        SectionCard(title = "Classes & Objects") {
            demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult ->
                result = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(
                modifier = Modifier.fillMaxWidth(),
                colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text(
                    text = "Result: $result",
                    modifier = Modifier.padding(16.dp),
                    fontSize = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text(
                text = title,
                fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
```

```
        }
    }
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    // Immutable variable (val)
    val immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types
    val byteValue: Byte = 127
    val shortValue: Short = 32767
    val intValue: Int = 2147483647
    val longValue: Long = 92233720368547758O7L
    val floatValue: Float = 3.14f
    val doubleValue: Double = 2.71828
    val charValue: Char = 'K'
    val booleanValue: Boolean = true
    val stringValue: String = "Hello Kotlin"
    val arrayValue: IntArray = intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue")
        Text("Float: $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}")
    }
}

@Composable
fun demonstrateOperators() {
    val a = 10
    val b = 3
```

```
    //Arithmetic operators
    valsum = a + b
    valdifference = a - b
    valproduct = a * b
    valquotient = a / b
    valremainder = a % b

    //Comparison operators
    valisEqual = a == b
    valisNotEqual = a != b
    valisGreater = a > b
    valisLess = a < b

    //Logical operators
    valx = true
    valy = false
    valandResult = x && y
    valorResult = x || y
    valnotResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text(" a + b = $sum")
        Text(" a - b = $difference")
        Text(" a * b = $product")
        Text(" a / b = $quotient")
        Text(" a % b = $remainder")

        Text("Comparison Operators:")
        Text(" a == b = $isEqual")
        Text(" a != b = $isNotEqual")
        Text(" a > b = $isGreater")
        Text(" a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text(" x && y = $andResult")
        Text(" x || y = $orResult")
        Text(" !x = $notResult")
    }
}

@Composable
fundemonstrateControlStatements() {
    valscore = 85
    valgrade = when {
        score >= 90 -> "A"
        score >= 80 -> "B"
```

```
        score >= 70 -> "C"
        score >= 60 -> "D"
        else -> "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5)
    val evenNumbers = mutableListOf<Int>()

    // For loop
    for (number in numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation
    var countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ")
            countdown--
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text(" Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text(" Numbers: ${numbers.joinToString()}")
        Text(" Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text(" Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello"
    val nullString: String? = null

    // Safe call operator
    val length1 = nullableString?.length

    // Elvis operator
```

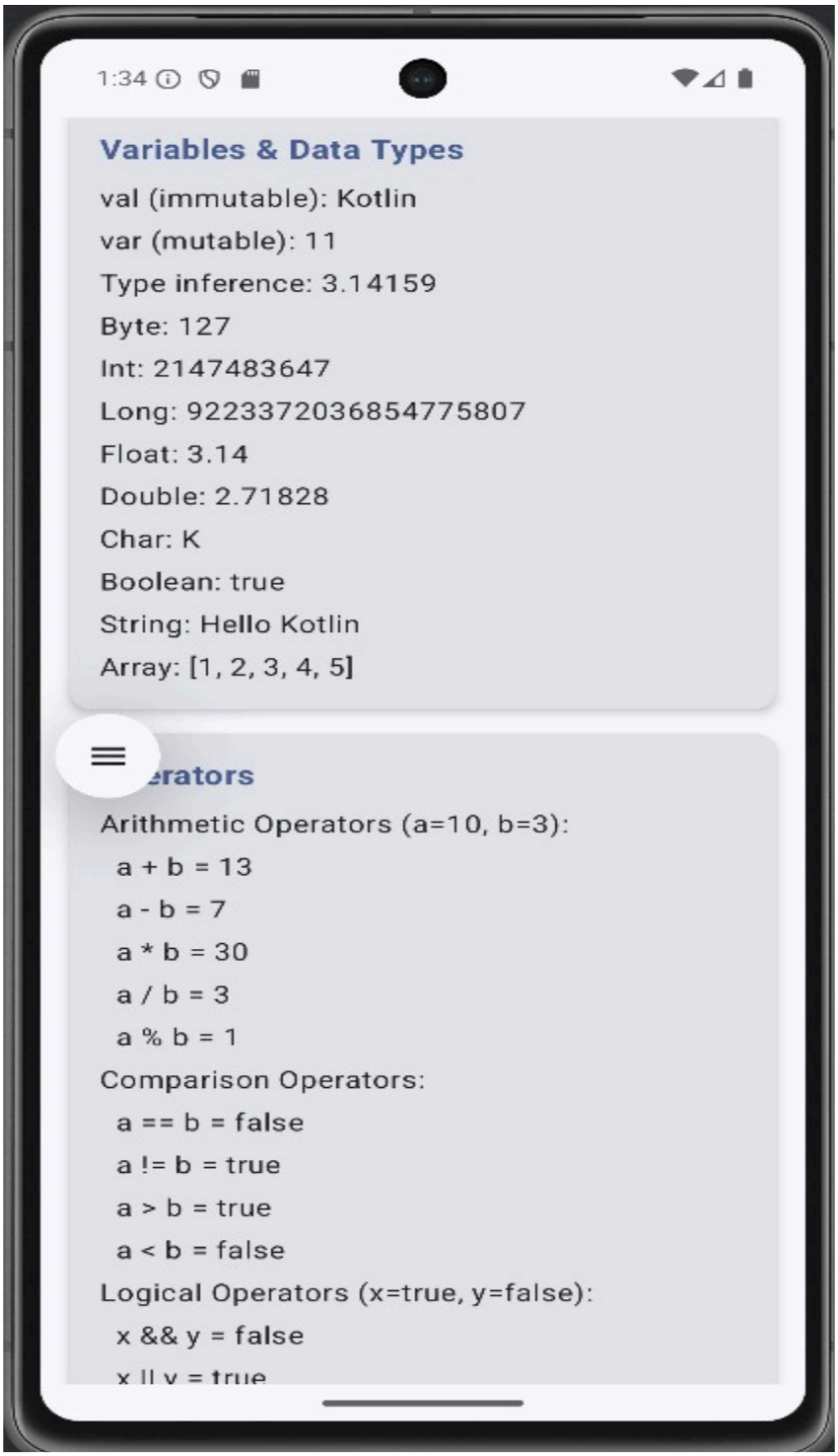
```
    val length2 = nullString?.length ?: 0

    // Non-null asserted call (use with caution)
    val length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function
    val result = nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() {
    // Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function
    fun String.isPalindrome(): Boolean {
        val cleaned = this.lowercase().filter { it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3)
    val greeting = greet("Kotlin")
    val sumVarargs = sumAll(1, 2, 3, 4, 5)
    val product = operate(4, 5, multiply)
    val palindrome = "Racecar".isPalindrome()
    val fact = factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Basic Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
```

```
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
    }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val priority = Priority.HIGH
    val config = AppConfig.getConfig()
    val area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy")
    val circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}")
    }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) {
var input1 by remember { mutableStateOf("") }
var input2 by remember { mutableStateOf("") }
var operation by remember { mutableStateOf("add") }

Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
        Text("Try it yourself!")

        OutlinedTextField(
            value = input1,
            onValueChange = { input1 = it },
            label = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField(
            value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
```

```
        )

        Row(
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            modifier = Modifier.fillMaxWidth()
        ) {
            RadioButton(
                selected = operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton(
                selected = operation == "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")

            RadioButton(
                selected = operation == "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button(
            onClick = {
                val num1 = input1.toIntOrNull() ?: 0
                val num2 = input2.toIntOrNull() ?: 0
                val result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2
                    "multiply" -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
```

```
        KotlinFundamentalsDemo()
    }
}
```

Output



Expt No : 2 **Date:**

Functions

Aim

Tounderstand and implement different types of functions in Kotlin.

Algorithm

1. Create a Kotlin file.
2. Define a function using fun keyword.
3. Pass parameters to the function.
4. Specify return type.
5. Call the function from main() or activity.
6. Implement default arguments.
7. Execute the program.

Coding

```
package com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview
import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!"
fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) {
```

```
 fun introduce(): String = "Hi, I'm $name and I'm $age years old"
}

data class User(val id: Int, val username: String, val email: String)

enum class Priority(val value: Int) {
    LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
    fun getDisplayName(): String = name.lowercase().replaceFirstChar { it.uppercase() }
}

object AppConfig {
    const val APP_NAME = "Kotlin Demo"
    const val VERSION = "1.0.0"
    fun getConfig(): String = "$APP_NAME v$VERSION"
}

class MathUtils {
    companion object {
        const val PI = 3.14159
        fun circleArea(radius: Double): Double = PI * radius * radius
    }
}

abstract class Animal(val name: String) {
    abstract fun makeSound(): String
    fun sleep(): String = "$name is sleeping"
}

class Dog(name: String) : Animal(name) {
    override fun makeSound(): String = "$name says: Woof!"
}

interface Drawable {
    fun draw(): String
}

class Circle(private val radius: Double) : Drawable {
    override fun draw(): String = "Drawing a circle with radius $radius"
}

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
```

```
                    modifier = Modifier.padding(innerPadding)
                )
            }
        }
    }
}

@Composable
fun KotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    var result by remember { mutableStateOf("") }

    Column(
        modifier = modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Kotlin Fundamentals Demo",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold
        )

        // Variables and Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        // Control Statements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") {
            demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
```

```
        }

        // Classes and Objects
        SectionCard(title = "Classes & Objects") {
            demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult ->
                result = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(
                modifier = Modifier.fillMaxWidth(),
                colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text(
                    text = "Result: $result",
                    modifier = Modifier.padding(16.dp),
                    fontSize = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text(
                text = title,
                fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
```

```
        }
    }
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    // Immutable variable (val)
    val immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types
    val byteValue: Byte = 127
    val shortValue: Short = 32767
    val intValue: Int = 2147483647
    val longValue: Long = 92233720368547758O7L
    val floatValue: Float = 3.14f
    val doubleValue: Double = 2.71828
    val charValue: Char = 'K'
    val booleanValue: Boolean = true
    val stringValue: String = "Hello Kotlin"
    val arrayValue: IntArray = intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val (immutable): $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type inference: $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue")
        Text("Float: $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}")
    }
}

@Composable
fun demonstrateOperators() {
    val a = 10
    val b = 3
```

```
    //Arithmetic operators
    valsum = a + b
    valdifference = a - b
    valproduct = a * b
    valquotient = a / b
    valremainder = a % b

    //Comparison operators
    valisEqual = a == b
    valisNotEqual = a != b
    valisGreater = a > b
    valisLess = a < b

    //Logical operators
    valx = true
    valy = false
    valandResult = x && y
    valorResult = x || y
    valnotResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text(" a + b = $sum")
        Text(" a - b = $difference")
        Text(" a * b = $product")
        Text(" a / b = $quotient")
        Text(" a % b = $remainder")

        Text("Comparison Operators:")
        Text(" a == b = $isEqual")
        Text(" a != b = $isNotEqual")
        Text(" a > b = $isGreater")
        Text(" a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text(" x && y = $andResult")
        Text(" x || y = $orResult")
        Text(" !x = $notResult")
    }
}

@Composable
fundemonstrateControlStatements() {
    valscore = 85
    valgrade = when {
        score >= 90 -> "A"
        score >= 80 -> "B"
```

```
        score >= 70 -> "C"
        score >= 60 -> "D"
        else -> "F"
    }

    val numbers = listOf(1, 2, 3, 4, 5)
    val evenNumbers = mutableListOf<Int>()

    // For loop
    for (number in numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation
    var countdown = 5
    val countdownResult = buildString {
        while (countdown > 0) {
            append("$countdown ")
            countdown--
        }
        append("Blast off!")
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text(" Score: $score -> Grade: $grade")

        Text("For Loop:")
        Text(" Numbers: ${numbers.joinToString()}")
        Text(" Even numbers: ${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text(" Countdown: $countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    val nullableString: String? = "Hello"
    val nullString: String? = null

    // Safe call operator
    val length1 = nullableString?.length

    // Elvis operator
```

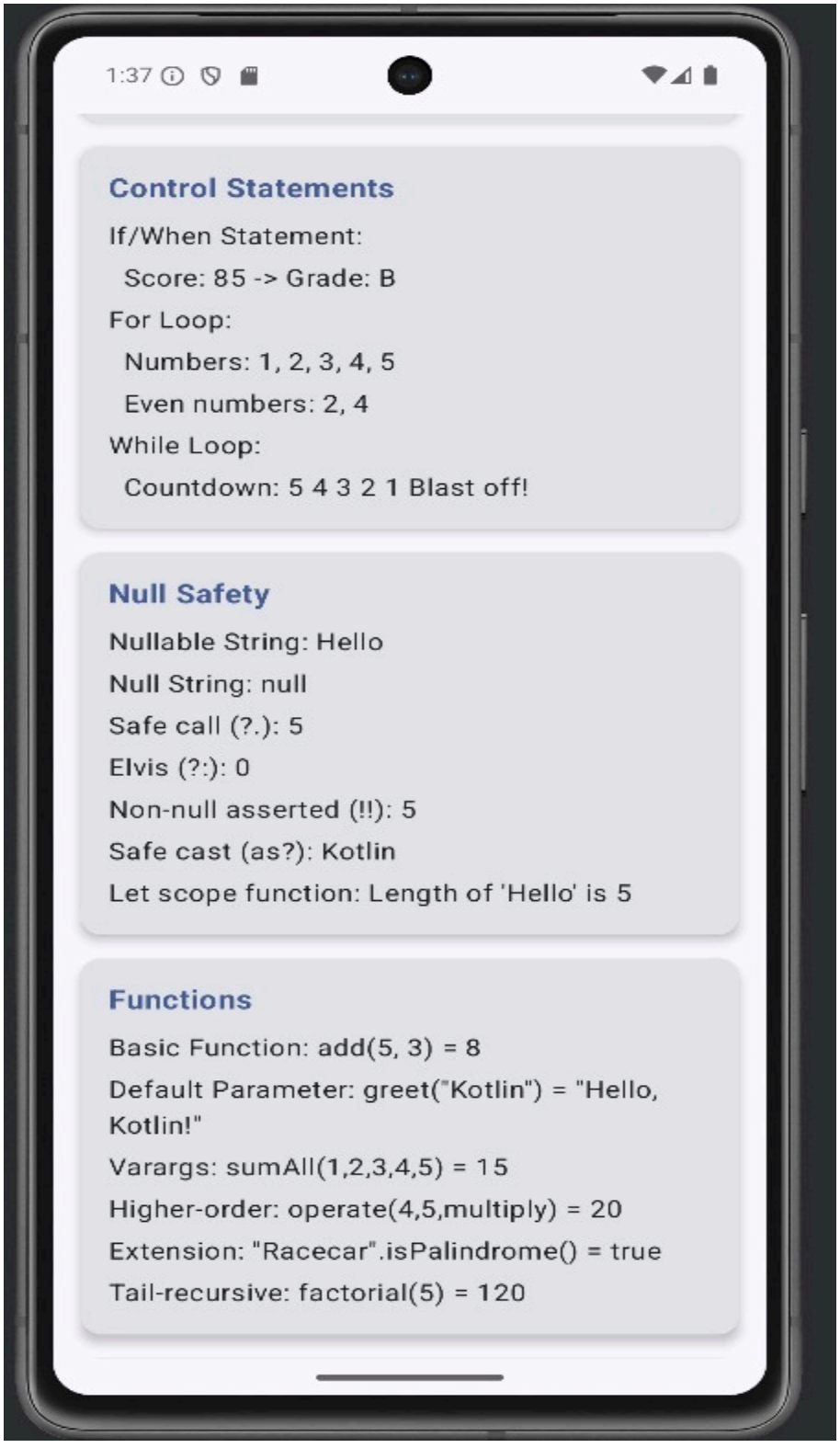
```
    val length2 = nullString?.length ?: 0

    // Non-null asserted call (use with caution)
    val length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function
    val result = nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() {
    // Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function
    fun String.isPalindrome(): Boolean {
        val cleaned = this.lowercase().filter { it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3)
    val greeting = greet("Kotlin")
    val sumVarargs = sumAll(1, 2, 3, 4, 5)
    val product = operate(4, 5, multiply)
    val palindrome = "Racecar".isPalindrome()
    val fact = factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Basic Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
```

```
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
    }
}

@Composable
fun demonstrateClassesAndObjects() {
    // Usage examples
    val person = Person("Alice", 25)
    val user = User(1, "john_doe", "john@example.com")
    val priority = Priority.HIGH
    val config = AppConfig.getConfig()
    val area = MathUtils.circleArea(5.0)
    val dog = Dog("Buddy")
    val circle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}")
    }
}

@Composable
fun InteractiveExample(onResult: (String) -> Unit) {
var input1 by remember { mutableStateOf("") }
var input2 by remember { mutableStateOf("") }
var operation by remember { mutableStateOf("add") }

Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
        Text("Try it yourself!")

        OutlinedTextField(
            value = input1,
            onValueChange = { input1 = it },
            label = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField(
            value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
```

```
        )

        Row(
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            modifier = Modifier.fillMaxWidth()
        ) {
            RadioButton(
                selected = operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton(
                selected = operation == "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")

            RadioButton(
                selected = operation == "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button(
            onClick = {
                val num1 = input1.toIntOrNull() ?: 0
                val num2 = input2.toIntOrNull() ?: 0
                val result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2
                    "multiply" -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
```

```
        KotlinFundamentalsDemo()
    }
}
```

Output



Expt No : 3 **Date:**

Classes and Objects

Aim

Toimplement object-oriented programming concepts using classes and objects in Kotlin.

Algorithm

1. Define a class using class keyword.
2. Declare properties inside the class.
3. Define member functions.
4. Create object using constructor.
5. Access class members.
6. Execute and verify output.

Coding

```
package com.example.basic1

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.compose.ui.tooling.preview.Preview
import com.example.basic1.ui.theme.Basic1Theme

// Function examples defined outside composable
fun add(a: Int, b: Int): Int = a + b
fun greet(name: String, greeting: String = "Hello"): String = "$greeting, $name!"
fun sumAll(vararg numbers: Int): Int = numbers.sum()
fun operate(a: Int, b: Int, operation: (Int, Int) -> Int): Int = operation(a, b)
tailrec fun factorial(n: Int, acc: Int = 1): Int = if (n <= 1) acc else factorial(n - 1, acc * n)

// Class examples defined outside composable
class Person(val name: String, var age: Int) {

    fun introduce(): String = "Hi, I'm $name and I'm $age years old"
}
```

```
dataclassUser(valid:Int,valusername: String, val email: String)

enumclassPriority(val value: Int) {
    LOW(1), MEDIUM(2), HIGH(3), URGENT(4);
    fungetDisplayName():String=name.lowercase().replaceFirstChar { it. uppercase() }
}

object AppConfig {
    constvalAPP_NAME="KotlinDemo"
    const val VERSION = "1.0.0"
    fungetConfig():String="$APP_NAME v$VERSION"
}

class MathUtils {
    companion object {
        const val PI = 3.14159
        funcircleArea(radius:Double):Double = PI * radius * radius
    }
}

abstractclassAnimal(valname:String) {
    abstract fun makeSound(): String
    funsleep():String="$nameissleeping"
}

classDog(name:String):Animal(name) {
    overridefunmakeSound():String= "$name says: Woof!"
}

interface Drawable {
    fun draw(): String
}

classCircle(privatevalradius:Double) : Drawable {
    overridefundraw():String="Drawing a circle with radius $radius"
}

classMainActivity:ComponentActivity() {
    overridefunonCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            Basic1Theme {
                Scaffold(modifier=Modifier.fillMaxSize()) { innerPadding ->
                    KotlinFundamentalsDemo(
                        modifier=Modifier.padding(innerPadding)
                    )
```

```
            }
        }
    }
}

@Composable
funKotlinFundamentalsDemo(modifier: Modifier = Modifier) {
    varresultbyremember { mutableStateOf("") }

    Column(
        modifier=modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text="Kotlin Fundamentals Demo",
            fontSize=24.sp,
            fontWeight= FontWeight.Bold
        )

        //Variablesand Data Types
        SectionCard(title = "Variables & Data Types") {
            demonstrateVariablesAndDataTypes()
        }

        // Operators
        SectionCard(title = "Operators") {
            demonstrateOperators()
        }

        //ControlStatements
        SectionCard(title = "Control Statements") {
            demonstrateControlStatements()
        }

        // Null Safety
        SectionCard(title = "Null Safety") {
            demonstrateNullSafety()
        }

        // Functions
        SectionCard(title = "Functions") {
            demonstrateFunctions()
        }
```

```
        // Classes and Objects
        SectionCard(title = "Classes & Objects") {
            demonstrateClassesAndObjects()
        }

        // Interactive Example
        SectionCard(title = "Interactive Example") {
            InteractiveExample { newResult ->
                result = newResult
            }
        }

        if (result.isNotEmpty()) {
            Card(
                modifier = Modifier.fillMaxWidth(),
                colors = CardDefaults.cardColors(
                    containerColor = MaterialTheme.colorScheme.primaryContainer
                )
            ) {
                Text(
                    text = "Result: $result",
                    modifier = Modifier.padding(16.dp),
                    fontSize = 16.sp
                )
            }
        }
    }
}

@Composable
fun SectionCard(title: String, content: @Composable () -> Unit) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
    ) {
        Column(
            modifier = Modifier.padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            Text(
                text = title,
                fontSize = 18.sp,
                fontWeight = FontWeight.Bold,
                color = MaterialTheme.colorScheme.primary
            )
            content()
        }
    }
```

```
}

@Composable
fun demonstrateVariablesAndDataTypes() {
    // Immutable variable (val)
    val immutableName: String = "Kotlin"

    // Mutable variable (var)
    var mutableAge: Int = 10
    mutableAge += 1

    // Type inference
    val inferredDouble = 3.14159

    // Different data types
    val byteValue: Byte = 127
    val shortValue: Short = 32767
    val intValue: Int = 2147483647
    val longValue: Long = 92233720368547758O7L
    val floatValue: Float = 3.14f
    val doubleValue: Double = 2.71828
    val charValue: Char = 'K'
    val booleanValue: Boolean = true
    val stringValue: String = "Hello Kotlin"
    val arrayValue: IntArray = intArrayOf(1, 2, 3, 4, 5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("val      (immutable):      $immutableName")
        Text("var (mutable): $mutableAge")
        Text("Type        inference:        $inferredDouble")
        Text("Byte: $byteValue")
        Text("Int: $intValue")
        Text("Long: $longValue")
        Text("Float: $floatValue")
        Text("Double: $doubleValue")
        Text("Char: $charValue")
        Text("Boolean: $booleanValue")
        Text("String: $stringValue")
        Text("Array: ${arrayValue.contentToString()}")
    }
}

@Composable
fun demonstrateOperators() {
    val a = 10
    val b = 3

    // Arithmetic operators
```

```
    val sum = a + b
    val difference = a - b
    val product = a * b
    val quotient = a / b
    val remainder = a % b

    // Comparison operators
    val isEqual = a == b
    val isNotEqual = a != b
    val isGreater = a > b
    val isLess = a < b

    // Logical operators
    val x = true
    val y = false
    val andResult = x && y
    val orResult = x || y
    val notResult = !x

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Arithmetic Operators (a=$a, b=$b):")
        Text(" a + b = $sum")
        Text(" a - b = $difference")
        Text(" a * b = $product")
        Text(" a / b = $quotient")
        Text(" a % b = $remainder")

        Text("Comparison Operators:")
        Text(" a == b = $isEqual")
        Text(" a != b = $isNotEqual")
        Text(" a > b = $isGreater")
        Text(" a < b = $isLess")

        Text("Logical Operators (x=$x, y=$y):")
        Text(" x && y = $andResult")
        Text(" x || y = $orResult")
        Text(" !x = $notResult")
    }
}

@Composable
fun demonstrateControlStatements() {
    val score = 85
    val grade = when {
        score >= 90 -> "A"
        score >= 80 -> "B"
        score >= 70 -> "C"
        score >= 60 -> "D"
```

```
        else ->"F"
    }

    val numbers = listOf(1,2,3,4,5)
    valevenNumbers=   mutableListOf<Int>()

    // For loop
    for (number in numbers) {
        if (number % 2 == 0) {
            evenNumbers.add(number)
        }
    }

    // While loop simulation
    var countdown = 5
    valcountdownResult=buildString{
        while (countdown > 0) {
            append("$countdown ")
            countdown--
        }
        append("Blast off!")
    }

    Column(verticalArrangement=Arrangement.spacedBy(4.dp)) {
        Text("If/When Statement:")
        Text("Score:$score->Grade:$grade")

        Text("For Loop:")
        Text("Numbers:${numbers.joinToString()}")
        Text("Evennumbers:${evenNumbers.joinToString()}")

        Text("While Loop:")
        Text("Countdown:$countdownResult")
    }
}

@Composable
fun demonstrateNullSafety() {
    // Nullable type
    valnullableString:String?="Hello"
    val nullString: String? = null

    // Safe call operator
    vallength1=nullableString?.length

    // Elvis operator
    vallength2=nullString?.length?:0
```

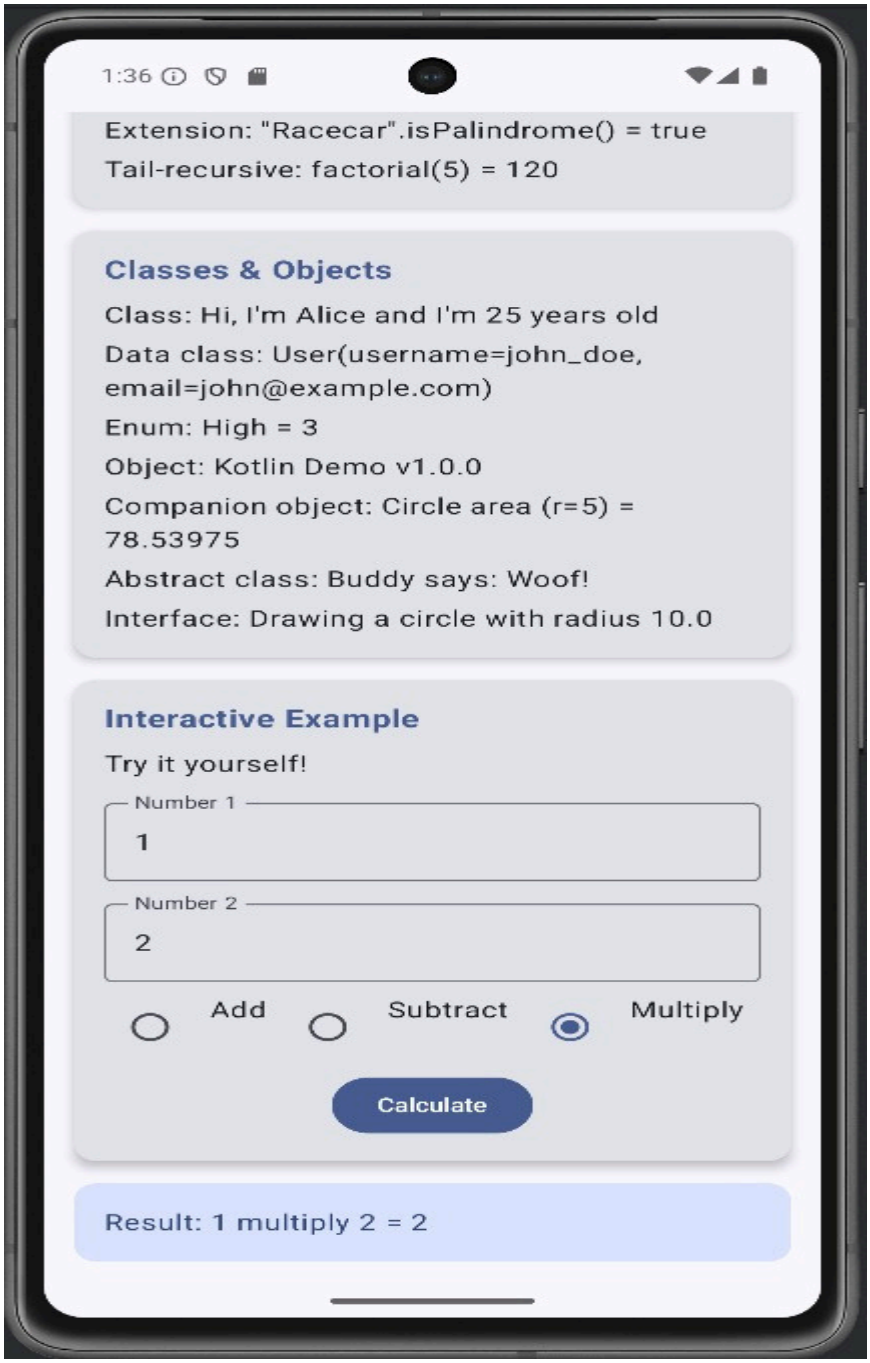
```
    // Non-null asserted call (use with caution)
    val length3 = nullableString!!.length

    // Safe cast
    val anyValue: Any = "Kotlin"
    val safeCast = anyValue as? String

    // Let scope function
    val result = nullableString?.let {
        "Length of '$it' is ${it.length}"
    }

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Nullable String: $nullableString")
        Text("Null String: $nullString")
        Text("Safe call (?.): $length1")
        Text("Elvis (?:): $length2")
        Text("Non-null asserted (!!): $length3")
        Text("Safe cast (as?): $safeCast")
        Text("Let scope function: $result")
    }
}

@Composable
fun demonstrateFunctions() {
    // Lambda function
    val multiply = { a: Int, b: Int -> a * b }

    // Extension function
    fun String.isPalindrome(): Boolean {
        val cleaned = this.lowercase().filter { it.isLetter() }
        return cleaned == cleaned.reversed()
    }

    val sum = add(5, 3)
    val greeting = greet("Kotlin")
    val sumVarargs = sumAll(1, 2, 3, 4, 5)
    val product = operate(4, 5, multiply)
    val palindrome = "Racecar".isPalindrome()
    val fact = factorial(5)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Basic Function: add(5, 3) = $sum")
        Text("Default Parameter: greet(\"Kotlin\") = \"$greeting\"")
        Text("Varargs: sumAll(1,2,3,4,5) = $sumVarargs")
        Text("Higher-order: operate(4,5,multiply) = $product")
        Text("Extension: \"Racecar\".isPalindrome() = $palindrome")
        Text("Tail-recursive: factorial(5) = $fact")
```

```
    }
}

@Composable
fundemonstrateClassesAndObjects() {
    //Usage examples
    valperson = Person("Alice", 25)
    valuser = User(1, "john_doe", "john@example.com")
    valpriority = Priority.HIGH
    valconfig = AppConfig.getConfig()
    valarea = MathUtils.circleArea(5.0)
    valdog = Dog("Buddy")
    valcircle = Circle(10.0)

    Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
        Text("Class: ${person.introduce()}")
        Text("Data class: User(username=${user.username}, email=${user.email})")
        Text("Enum: ${priority.getDisplayName()} = ${priority.value}")
        Text("Object: $config")
        Text("Companion object: Circle area (r=5) = $area")
        Text("Abstract class: ${dog.makeSound()}")
        Text("Interface: ${circle.draw()}")
    }
}

@Composable
funInteractiveExample(onResult: (String) -> Unit) {
varinput1 by remember { mutableStateOf("") }
varinput2 by remember { mutableStateOf("") }
varoperation by remember { mutableStateOf("add") }

Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
        Text("Try it yourself!")

        OutlinedTextField(
            value = input1,
            onValueChange = { input1 = it },
            label = { Text("Number 1") },
            modifier = Modifier.fillMaxWidth()
        )

        OutlinedTextField(
            value = input2,
            onValueChange = { input2 = it },
            label = { Text("Number 2") },
            modifier = Modifier.fillMaxWidth()
        )
```

```
        Row(
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            modifier = Modifier.fillMaxWidth()
        ) {
            RadioButton(
                selected = operation == "add",
                onClick = { operation = "add" }
            )
            Text("Add")

            RadioButton(
                selected = operation == "subtract",
                onClick = { operation = "subtract" }
            )
            Text("Subtract")

            RadioButton(
                selected = operation == "multiply",
                onClick = { operation = "multiply" }
            )
            Text("Multiply")
        }

        Button(
            onClick = {
                val num1 = input1.toIntOrNull() ?: 0
                val num2 = input2.toIntOrNull() ?: 0
                val result = when (operation) {
                    "add" -> num1 + num2
                    "subtract" -> num1 - num2
                    "multiply" -> num1 * num2
                    else -> 0
                }
                onResult("$num1 $operation $num2 = $result")
            },
            modifier = Modifier.align(Alignment.CenterHorizontally)
        ) {
            Text("Calculate")
        }
    }
}

@Preview(showBackground = true)
@Composable
fun KotlinFundamentalsDemoPreview() {
    Basic1Theme {
        KotlinFundamentalsDemo()
    }
```

}

Output


1:36
Extension: "Racecar".isPalindrome() = true
Tail-recursive: factorial(5) = 120
Classes & Objects
Class: Hi, I'm Alice and I'm 25 years old
Data class: User(username=john_doe, email=john@example.com)
Enum: High = 3
Object: Kotlin Demo v1.0.0
Companion object: Circle area (r=5) = 78.53975
Abstract class: Buddy says: Woof!
Interface: Drawing a circle with radius 10.0
Interactive Example
Try it yourself!
Number 1
1
Number 2
2
Add
Subtract
Multiply
Calculate
Result: 1 multiply 2 = 2

Expt No : 4 **Date:**

Build FirstAndroid App

Aim
Tocreate and execute a basic Android application using Kotlin.

Algorithm
1. Open Android Studio.
2. Create new project → Empty Activity.
3. Design UI using XML layout.
4. Write Kotlin logic in Activity file.
5. Build the project.
6. Run app on emulator/device.

AndroidManifest.xml
```
<?xmlversion="1.0"encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
<application
android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:supportsRtl="true"
android:theme="@style/Theme.GraphicalPrimitives"
tools:targetApi="31">
<activity
android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

activity_main.xml
```
<?xmlversion="1.0" encoding="utf-8"?>
```

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<org.rajalakshmi.graphicalprimitives.SampleCanvas
android:layout_width="match_parent"
android:layout_height="match_parent">
</org.rajalakshmi.graphicalprimitives.SampleCanvas>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.kt

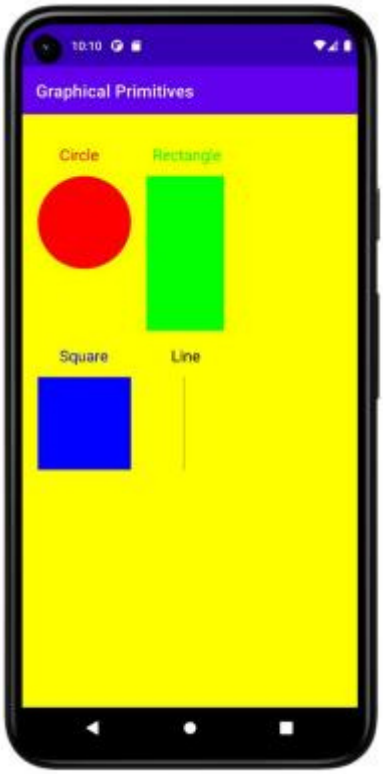
```
package org.rajalakshmi.graphicalprimitives
import    androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
}
}
```

SampleCanvas.kt

```
package org.rajalakshmi.graphicalprimitives
import android.content.Context
import android.graphics.Canvas
import android.graphics.Color
import android.graphics.Paint
import android.util.AttributeSet
import android.view.View
class SampleCanvas @JvmOverloads constructor(
context: Context, attrs: AttributeSet? = null, defStyleAttr: Int = 0
) : View(context, attrs, defStyleAttr) {
override fun onDraw(canvas: Canvas?) {
super.onDraw(canvas)
val paint : Paint = Paint()
paint.setColor(Color.YELLOW)
canvas?.drawPaint(paint)
paint.setTextSize(50f);
paint.setColor(Color.RED);
canvas?.drawText("Circle", 120f, 150f, paint);
canvas?.drawCircle(200f, 350f, 150f, paint);
paint.setColor(Color.GREEN);

canvas?.drawText("Rectangle", 420f, 150f, paint);
canvas?.drawRect(400f, 200f, 650f, 700f, paint);
```

```
paint.setColor(Color.BLUE);
canvas?.drawText("Square", 120f, 800f, paint);
canvas?.drawRect(50f, 850f, 350f, 1150f, paint);
paint.setColor(Color.BLACK);
canvas?.drawText("Line", 480f, 800f, paint);
canvas?.drawLine(520f, 850f, 520f, 1150f, paint);
}
}
```

Output



Expt No : 5 **Date:**

Layouts

Aim
Todesign user interfaces using Android layout managers.

Algorithm
1. Open XML layout file.
2. Select layout type.
3. Add UI components.
4. Set layout parameters.
5. Preview and run application.

Mainactivity.kt

```
package com.example.androidlayoutmanagers

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.rememberScrollState
import androidx.compose.foundation.verticalScroll
import androidx.compose.material3.*
import androidx.compose.material3.ExperimentalMaterial3Api
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import com.example.androidlayoutmanagers.ui.theme.AndroidLayoutManagersTheme

class MainActivity : ComponentActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            AndroidLayoutManagersTheme {
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    LayoutManagersApp()
```

```
            }
        }
    }
}

@Composable
fun LayoutManagersApp() {
    var selectedDemo by remember { mutableStateOf<DemoType?>(null) }

    selectedDemo?.let { demo ->
        DemoScreen(demo = demo, onBack = { selectedDemo = null })
    } ?: MainScreen(onDemoSelected = { selectedDemo = it })
}

@Composable
fun MainScreen(onDemoSelected: (DemoType) -> Unit) {
    Column(
        modifier = Modifier
            .fillMaxSize()
            .verticalScroll(rememberScrollState())
            .padding(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        Text(
            text = "Android Layout Managers",
            fontSize = 32.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.align(Alignment.CenterHorizontally)
        )

        Text(
            text = "Explore different layout managers in Jetpack Compose",
            fontSize = 16.sp,
            color = MaterialTheme.colorScheme.onSurfaceVariant,
            modifier = Modifier.align(Alignment.CenterHorizontally)
        )

        Spacer(modifier = Modifier.height(16.dp))

        DemoType.values().forEach { demo ->
            Card(
                modifier = Modifier.fillMaxWidth(),
                elevation = CardDefaults.cardElevation(defaultElevation = 4.dp),
                onClick = { onDemoSelected(demo) }
            ) {
                Column(
                    modifier = Modifier
```

```
                .fillMaxWidth()
                .padding(16.dp)
            ) {
                Text(
                    text = demo.title,
                    fontSize = 20.sp,
                    fontWeight = FontWeight.Bold
                )
                Spacer(modifier = Modifier.height(4.dp))
                Text(
                    text = demo.description,
                    fontSize = 14.sp,
                    color = MaterialTheme.colorScheme.onSurfaceVariant
                )
            }
        }
    }
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun DemoScreen(demo: DemoType, onBack: () -> Unit) {
    Column(modifier = Modifier.fillMaxSize()) {
        TopAppBar(
            title = { Text(demo.title) },
            navigationIcon = {
                IconButton(onClick = onBack) {
                    Text("← Back")
                }
            }
        )

        Box(modifier = Modifier.fillMaxSize()) {
            when (demo) {
                DemoType.COLUMN -> ColumnExample()
                DemoType.ROW -> RowExample()
                DemoType.BOX -> BoxExample()
                DemoType.LAZY_COLUMN -> LazyColumnExample()
                DemoType.LAZY_ROW -> LazyRowExample()
                DemoType.MIXED_LAZY -> MixedLazyLayoutExample()
                DemoType.CONSTRAINT_BASIC -> ConstraintLayoutBasicExample()
                DemoType.CONSTRAINT_COMPLEX -> ConstraintLayoutComplexExample()
                DemoType.CONSTRAINT_GUIDELINES ->
ConstraintLayoutGuidelinesExample()
                DemoType.GRID_FIXED -> LazyVerticalGridExample()
                DemoType.GRID_ADAPTIVE -> AdaptiveGridExample()
                DemoType.GRID_VARIABLE -> VariableGridExample()
```

```
            DemoType.PRODUCT_GRID -> ProductGridExample()
        }
    }
  }
}

enumclassDemoType(
  valtitle:String,
  valdescription: String
) {
  COLUMN("Column Layout", "Vertical arrangement of components"),
  ROW("Row Layout", "Horizontal arrangement of components"),
  BOX("Box Layout", "Stacked components with positioning"),
  LAZY_COLUMN("LazyColumn", "Efficient vertical scrolling lists"),
  LAZY_ROW("LazyRow", "Efficient horizontal scrolling lists"),
  MIXED_LAZY("Mixed Lazy Layout", "Combination of LazyColumn and LazyRow"),
  CONSTRAINT_BASIC("ConstraintLayout Basic", "Basic constraint positioning"),
  CONSTRAINT_COMPLEX("ConstraintLayout Complex", "Complex profile layout"),
  CONSTRAINT_GUIDELINES("ConstraintLayout Guidelines", "Layout with
guidelines"),
  GRID_FIXED("Fixed Grid", "Fixed number of columns"),
  GRID_ADAPTIVE("Adaptive Grid", "Adaptive column sizing"),
  GRID_VARIABLE("Variable Grid", "Variable item sizes"),
  PRODUCT_GRID("Product Grid", "Real-world product grid")
}
```

settings.gradle.kts

```
pluginManagement {
    repositories {
        google {
            content {
                includeGroupByRegex("com\\.android.*")
                includeGroupByRegex("com\\.google.*")
                includeGroupByRegex("androidx.*")
            }
        }
        mavenCentral()
        gradlePluginPortal()
    }
}
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
    repositories {
    google()
    mavenCentral()
```

```
    }
}
rootProjectname
                .          = "Android layout managers"
include(":app")
```

ConstraintsLayoutExamples.kts

```
package com.example.androidlayoutmanagers
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp
import androidx.constraintlayout.compose.ConstraintLayout
import androidx.constraintlayout.compose.ConstraintSet
import androidx.constraintlayout.compose.Dimension

@Composable
fun ConstraintLayoutBasicExample() {
    ConstraintLayout(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ) {
        val (title, subtitle, button, image) = createRefs()

        Text(
            text = "ConstraintLayout Basic",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.constrainAs(title) {
                top.linkTo(parent.top, margin = 16.dp)
```

```
            start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    )

    Text(
        text="Positioning elements with constraints",
        fontSize=16.sp,
        color=MaterialTheme.colorScheme.onSurfaceVariant,
        modifier=Modifier.constrainAs(subtitle) {
            top.linkTo(title.bottom, margin = 8.dp)
            start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    )

    Button(
        onClick={},
        modifier=Modifier.constrainAs(button) {
            bottom.linkTo(parent.bottom, margin = 32.dp)
            start.linkTo(parent.start)
            end.linkTo(parent.end)
        }
    ) {
        Text("Centered Button")
    }
    Box(

        modifier=Modifier
            .size(100.dp)
            .background(
                Color.Blue,
                RoundedCornerShape(12.dp)
            )
            .constrainAs(image) {
                top.linkTo(subtitle.bottom, margin = 24.dp)
                bottom.linkTo(button.top, margin = 24.dp)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            },
        contentAlignment = Alignment.Center
    ) {
        Text(
            text="Center Box",
            color=Color.White,
            fontWeight = FontWeight.Bold
        )
    }
```

```
    }

}

@Composable
fun ConstraintLayoutComplexExample() {
    ConstraintLayout(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ){
        val(header,profilePic,userName, userEmail, statsContainer,
            bioText,actionButton, settingsButton) = createRefs()

        valbarrier=createEndBarrier(profilePic, userName, userEmail)

        Text(
            text="ProfileLayout",
            fontSize =20.sp,
            fontWeight=FontWeight.Bold,
            modifier=Modifier.constrainAs(header) {
                top.linkTo(parent.top)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            }
        )

        Box(
            modifier = Modifier
                .size(80.dp)
                .background(
                    Color.Green,
                    RoundedCornerShape(40.dp)
                )
                .constrainAs(profilePic) {
                    top.linkTo(header.bottom, margin = 16.dp)
                    start.linkTo(parent.start)
                },
            contentAlignment=Alignment.Center
        ){
            Text(
                text ="PIC",
                color=Color.White,
                fontWeight=FontWeight.Bold
            )
        }

        Text(
            text = "John Doe",
```

```
            fontSize= 18.sp,
            fontWeight = FontWeight.Bold,
            modifier= Modifier.constrainAs(userName) {
                start.linkTo(profilePic.end, margin = 16.dp)
                top.linkTo(profilePic.top)
            }
        )

        Text(
            text="john.doe@example.com",
            fontSize= 14.sp,
            color=MaterialTheme.colorScheme.onSurfaceVariant,
            modifier= Modifier.constrainAs(userEmail) {
                start.linkTo(profilePic.end, margin = 16.dp)
                top.linkTo(userName.bottom, margin = 4.dp)
            }
        )

        Box(
            modifier= Modifier
                .constrainAs(statsContainer) {
                    top.linkTo(profilePic.bottom, margin = 16.dp)
                    start.linkTo(parent.start)
                    end.linkTo(parent.end)
                    width = Dimension.fillToConstraints
                }
        ) {
            Row(
                modifier = Modifier.fillMaxWidth(),
                horizontalArrangement = Arrangement.SpaceEvenly
            ) {
                StatItem("Posts", "150", Modifier)
                StatItem("Followers", "2.5K", Modifier)
                StatItem("Following", "500", Modifier)
            }
        }

        Text(
            text= "Passionate developer and Android enthusiast. Love creating beautiful and
functionalmobile applications.",
            fontSize = 14.sp,
            modifier = Modifier.constrainAs(bioText) {
                top.linkTo(statsContainer.bottom, margin = 16.dp)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
                width = Dimension.fillToConstraints
            }
        )
```

```
        Button(
            onClick = { },
            modifier = Modifier.constrainAs(actionButton) {
            top.linkTo(bioText.bottom, margin = 16.dp)
            start.linkTo(parent.start)
            end.linkTo(settingsButton.start, margin = 8.dp)
            width = Dimension.fillToConstraints
            }
        ) {Text("Follow")

        }
        OutlinedButton(
        ,
            onClick =  { }
            modifier=Modifier.constrainAs(settingsButton) {
                top.linkTo(bioText.bottom, margin = 16.dp)
                end.linkTo(parent.end)
                width=Dimension.value(120.dp)
            }
        ) {
            Text("Message")
        }
    }
}

@Composable
privatefunStatItem(label: String, value: String, modifier: Modifier) {
    Column(
        horizontalAlignment = Alignment.CenterHorizontally,
        modifier=modifier
    ) {
        Text(
            text =value,
            fontSize=18.sp,
            fontWeight=FontWeight.Bold
        )
        Text(
            text =label,
            fontSize=12.sp,
            color=MaterialTheme.colorScheme.onSurfaceVariant
        )
    }
}

@Composable
funConstraintLayoutGuidelinesExample() {
    ConstraintLayout(
```

```
        modifier=Modifier
            .fillMaxSize()
            .padding(16.dp)
    ) {
        val(title,leftCard, centerCard, rightCard, bottomBar) = createRefs()

        valverticalGuideline = createGuidelineFromStart(0.33f)
        valverticalGuideline2 = createGuidelineFromEnd(0.33f)
        valhorizontalGuideline = createGuidelineFromTop(0.4f)

        Text(
            text="Guidelines Layout",
            fontSize=20.sp,
            fontWeight=FontWeight.Bold,
            modifier=Modifier.constrainAs(title) {
                top.linkTo(parent.top)
                start.linkTo(parent.start)
                end.linkTo(parent.end)
            }
        )

        Card(
            modifier=Modifier
                .fillMaxHeight()
                .constrainAs(leftCard) {
                    top.linkTo(title.bottom, margin = 16.dp)
                    bottom.linkTo(bottomBar.top, margin = 16.dp)
                    start.linkTo(parent.start)
                    end.linkTo(verticalGuideline)
                },
            colors=CardDefaults.cardColors(
                containerColor = Color.Red.copy(alpha = 0.7f)
            )
        ) {
            Box(
                contentAlignment = Alignment.Center,
                modifier=Modifier.fillMaxSize()
            ) {
                Text(
                    text="Left\n33%",
                    color=Color.White,
                    fontWeight = FontWeight.Bold
                )
            }
        }

        Card(
            modifier = Modifier
```

```
            .fillMaxHeight()
            .constrainAs(centerCard) {
                top.linkTo(title.bottom, margin = 16.dp)
                bottom.linkTo(bottomBar.top, margin = 16.dp)
                start.linkTo(verticalGuideline)
                end.linkTo(verticalGuideline2)
            },
        colors= CardDefaults.cardColors(
            containerColor = Color.Green.copy(alpha = 0.7f)
        )
    ) {
        Box(
            contentAlignment = Alignment.Center,
            modifier = Modifier.fillMaxSize()
        ) {
            Text(
                text = "Center\n34%",
                color = Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }

    Card(
        modifier = Modifier
            .fillMaxHeight()
            .constrainAs(rightCard) {
                top.linkTo(title.bottom, margin = 16.dp)
                bottom.linkTo(bottomBar.top, margin = 16.dp)
                start.linkTo(verticalGuideline2)
                end.linkTo(parent.end)
            },
        colors= CardDefaults.cardColors(
            containerColor = Color.Blue.copy(alpha = 0.7f)
        )
    ) {
        Box(
            contentAlignment = Alignment.Center,
            modifier = Modifier.fillMaxSize()
        ) {
            Text(
                text = "Right\n33%",
                color = Color.White,
                fontWeight = FontWeight.Bold
            )
        }
    }
```

```
        Card(
            modifier = Modifier
                .height(60.dp)
                .constrainAs(bottomBar) {
                    bottom.linkTo(parent.bottom)
                    start.linkTo(parent.start)
                    end.linkTo(parent.end)
                },
            colors= CardDefaults.cardColors(
                containerColor = MaterialTheme.colorScheme.surfaceVariant
            )
        ) {
            Box(
                contentAlignment = Alignment.Center,
                modifier = Modifier.fillMaxSize()
            ) {
                Text(
                    text = "Bottom Navigation Bar",
                    fontWeight = FontWeight.Medium
                )
            }
        }
    }
}
```

GridLayoutExamples.kt

```
package com.example.androidlayoutmanagers
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.grid.GridCells
import androidx.compose.foundation.lazy.grid.LazyVerticalGrid
import androidx.compose.foundation.lazy.grid.items
import androidx.compose.foundation.lazy.grid.rememberLazyGridState
import androidx.compose.foundation.shape.RoundedCornerShape
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

data class GridItem(val id: Int, val title: String, val color: Color)

@Composable
```

```
fun LazyVerticalGridExample() {
    val items= remember {
        (1..50).map { index ->
            valcolors = listOf(
                Color.Red, Color.Blue, Color.Green, Color.Yellow,
                Color.Magenta, Color.Cyan, Color.Gray, Color(0xFFFF6B35)
            )
            GridItem(
                id= index,
                title = "Item $index",
                color = colors[index % colors.size]
            )
        }
    }

    valgridState = rememberLazyGridState()

    Column(
        modifier = Modifier.fillMaxSize()
    ) {
        Text(
            text = "LazyVerticalGrid - 2 Columns",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.padding(16.dp)
        )

        LazyVerticalGrid(
            columns = GridCells.Fixed(2),
            modifier = Modifier.fillMaxSize(),
            state = gridState,
            contentPadding = PaddingValues(16.dp),
            horizontalArrangement = Arrangement.spacedBy(12.dp),
            verticalArrangement = Arrangement.spacedBy(12.dp)
        ) {
            items(items) { item ->
                Card(
                    modifier = Modifier
                    .fillMaxWidth()
                    .aspectRatio(1f),
                    elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
                ) {
                    Box(
                        modifier = Modifier
                            .fillMaxSize()
                            .background(item.color.copy(alpha = 0.8f)),
                        contentAlignment = Alignment.Center
                    ){
```

```
                    Text(
                        text = item.title,
                        color = Color.White,
                        fontWeight = FontWeight.Bold,
                        fontSize = 16.sp
                    )
                }
            }
        }
    }
}

@Composable
fun AdaptiveGridExample() {
    val items= remember {
        (1..30).map { index ->
            valcolors = listOf(
                Color(0xFF6200EE), Color(0xFF03DAC6), Color(0xFFCF6679),
                Color(0xFF3700B3), Color(0xFF018786), Color(0xFFBB86FC)
            )
            GridItem(
                id= index,
                title = "Adaptive $index",
                color = colors[index % colors.size]
            )
        }
    }
    Column(

        modifier = Modifier.fillMaxSize()
    ) {
        Text(
            text = "Adaptive Grid - Min 120dp",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.padding(16.dp)
        )

        LazyVerticalGrid(
            columns = GridCells.Adaptive(minSize = 120.dp),
            modifier = Modifier.fillMaxSize(),
            contentPadding = PaddingValues(16.dp),
            horizontalArrangement = Arrangement.spacedBy(8.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ){
            items(items) { item ->
                Card(
```

```
                modifier = Modifier
                .fillMaxWidth()
                .height(100.dp),
                elevation = CardDefaults.cardElevation(defaultElevation = 3.dp)
            ) {
                Box(
```

LazyLayoutExamples.kt

```
packagecom.example.androidlayoutmanagers

importandroidx.compose.foundation.background
importandroidx.compose.foundation.layout.*
importandroidx.compose.foundation.lazy.LazyColumn
importandroidx.compose.foundation.lazy.LazyRow
importandroidx.compose.foundation.lazy.items
importandroidx.compose.foundation.lazy.rememberLazyListState
importandroidx.compose.foundation.shape.RoundedCornerShape
importandroidx.compose.material3.*
importandroidx.compose.runtime.*
importandroidx.compose.ui.Alignment
importandroidx.compose.ui.Modifier
importandroidx.compose.ui.graphics.Color
importandroidx.compose.ui.text.font.FontWeight
importandroidx.compose.ui.unit.dp
importandroidx.compose.ui.unit.sp

dataclassItem(valid: Int, val title: String, val description: String)

@Composable
funLazyColumnExample() {
    valitems=remember {
        (1..50).map { index ->
            Item(
                id=index,
                title="Item $index",
                description = "This is the description for item number $index"
            )
        }
    }

    vallistState = rememberLazyListState()

    Column(
        modifier = Modifier.fillMaxSize()
```

```
    ) {
        Text(
            text = "LazyColumn - ${items.size} items",
            fontSize = 24.sp,
            fontWeight = FontWeight.Bold,
            modifier = Modifier.padding(16.dp)
        )

        LazyColumn(
            modifier = Modifier.fillMaxSize(),
            state = listState,
            contentPadding = PaddingValues(16.dp),
            verticalArrangement = Arrangement.spacedBy(8.dp)
        ) {
            items(items) { item ->
                Card(
                    modifier = Modifier
                        .fillMaxWidth(),
                    elevation = CardDefaults.cardElevation(defaultElevation = 4.dp)
                ) {
                    Column(
                        modifier = Modifier
                        .fillMaxWidth()
                        .padding(16.dp)
                    ) {
                        Text(
                            text = item.title,
                            fontSize = 18.sp,
                            fontWeight = FontWeight.Bold,
                            color = MaterialTheme.colorScheme.primary
                        )
                        Spacer(modifier = Modifier.height(4.dp))
                        Text(
                            text = item.description,
                            fontSize = 14.sp,
                            color = MaterialTheme.colorScheme.onSurfaceVariant
                        )
                    }
                }
            }
        }
    }
}

@Composable
fun LazyRowExample() {
    val categories = remember {
        listOf(
```

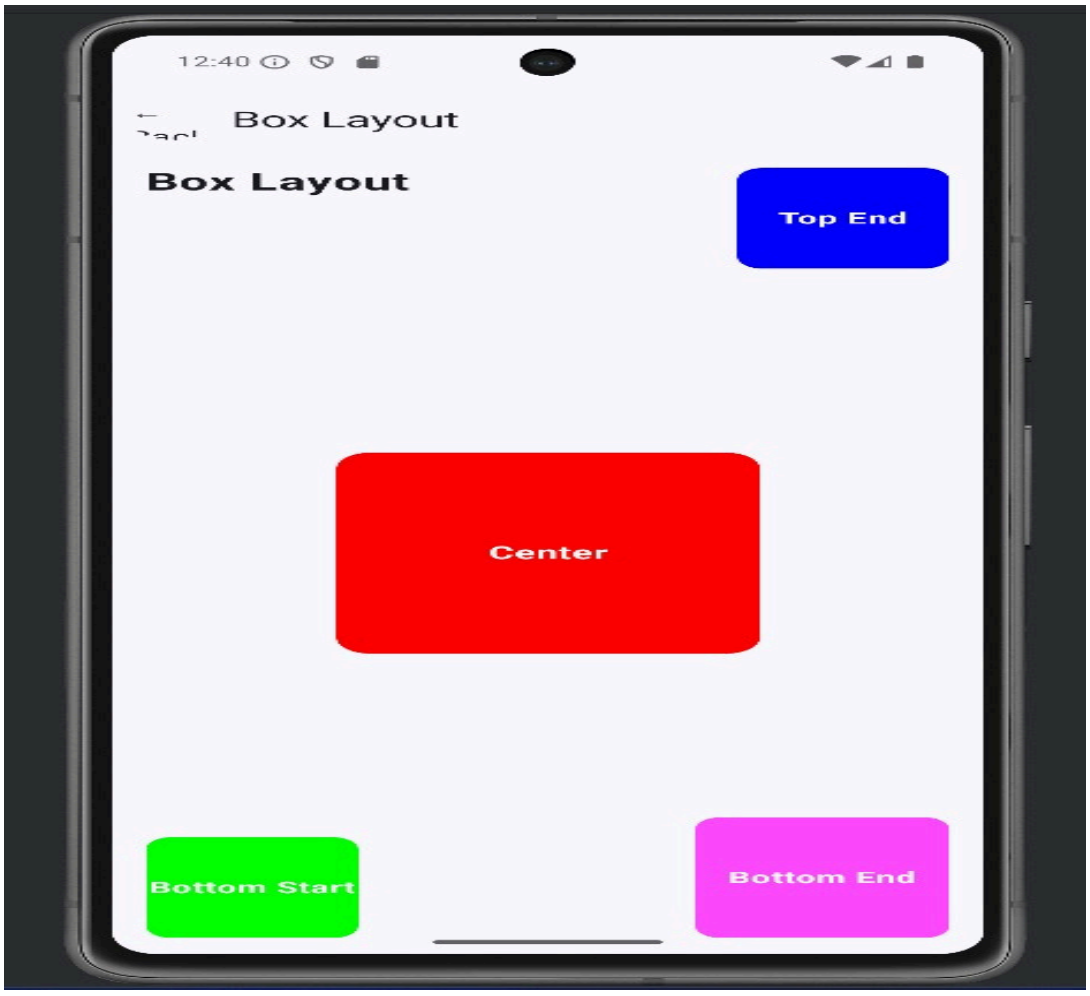
```
        "Technology", "Sports", "Music", "Movies", "Books",
        "Travel", "Food", "Fashion", "Gaming", "Art",
        "Science", "History", "Nature", "Photography", "Dance"
    )
}

vallistState = rememberLazyListState()

Column(
    modifier = Modifier
        .fillMaxSize()
        .padding(16.dp)
) {
    Text(
        text = "LazyRow - Horizontal Scrolling",
        fontSize = 24.sp,
        fontWeight = FontWeight.Bold,
        modifier = Modifier.padding(bottom = 16.dp)
    )

    LazyRow(
        state= listState,
        horizontalArrangement=Arrangement.spacedBy(12.dp),
        contentPadding=PaddingValues(horizontal = 16.dp)
    ) {
        items(categories) { category ->
            Card(
                modifier = Modifier
                    .width(120.dp)
                    .height(80.dp),
                shape=RoundedCornerShape(12.dp),
                elevation=CardDefaults.cardElevation(defaultElevation = 6.dp)
            ) {
                Box(
                    modifier = Modifier
                        .fillMaxSize()
                        .background(
                            Color(
                                red=(0.3f+(category.hashCode() % 100) / 200f),
                                green=(0.5f+(category.hashCode() % 150) / 300f),
                                blue=(0.7f+(category.hashCode() % 200) / 400f)
                            )
                        ),
                    contentAlignment=Alignment.Center
                ) {
                    Text(
                        text = category,
                        color = Color.White,
```

```
                    fontWeight = FontWeight.Bold,
                    modifier = Modifier.padding(8.dp)
                )
            }
        }
    }
}

Spacer(modifier = Modifier.height(24.dp))

Text(
    text = "Featured Items",
    fontSize = 20.sp,
    fontWeight = FontWeight.Bold,
    modifier = Modifier.padding(bottom = 12.dp)
)

valfeaturedItems = remember {
    (1..20).map { index ->
        "Featured $index"
    }
}

LazyRow(
    horizontalArrangement = Arrangement.spacedBy(8.dp),
    contentPadding = PaddingValues(horizontal = 16.dp)
) {
    items(featuredItems) { item ->
        Card(
            modifier = Modifier
            .width(150.dp)
            .height(100.dp),
            colors= CardDefaults.cardColors(
                containerColor = MaterialTheme.colorScheme.surfaceVariant
            )
        ) {
            Box(
                contentAlignment = Alignment.Center
            ) {
                Text(
                    text = item,
                    fontWeight = FontWeight.Medium,
                    modifier = Modifier.padding(12.dp)
                )
            }
        }
    }
}
```

```
    }
}

@Composable
funMixedLazyLayoutExample() {
    valsections = remember {
        listOf(
            "Header" to listOf("Welcome to Lazy Layouts"),
            "Categories" to listOf("Tech", "Sports", "Music", "Movies", "Books"),
            "Popular Items" to (1..10).map { "Popular Item $it" },
            "Recent Items" to (1..15).map { "Recent Item $it" }
        )
    }
    LazyColumn(

        modifier = Modifier.fillMaxSize(),
        contentPadding = PaddingValues(16.dp),
        verticalArrangement = Arrangement.spacedBy(16.dp)
    ) {
        sections.forEach { (sectionTitle, items) ->
            item{
                Text(
                    text= sectionTitle,
                    fontSize = 20.sp,
                    fontWeight = FontWeight.Bold,
                    modifier = Modifier.padding(vertical = 8.dp)
                )
            }
            if(sectionTitle == "Categories") {

                item{
                    LazyRow(
                        horizontalArrangement = Arrangement.spacedBy(8.dp)
                    ) {
                        items(items){ category ->
                            Card(
                                modifier= Modifier
                                    .width(100.dp)
                                    .height(60.dp),
                                colors=CardDefaults.cardColors(
                                    containerColor = MaterialTheme.colorScheme.primaryContainer
                                )
                            ) {
                                Box(
                                    contentAlignment = Alignment.Center
                                ) {
                                    Text(
                                        text= category,
```

```
                        fontWeight = FontWeight.Medium,
                        color = MaterialTheme.colorScheme.onPrimaryContainer
                    )
                }
            }
        }
    }
}else {
    items(items) { item ->
        Card(
            modifier = Modifier.fillMaxWidth(),
            elevation = CardDefaults.cardElevation(defaultElevation = 2.dp)
        ) {
            Text(
                text = item,
                modifier = Modifier.padding(16.dp)
            )
        }
    }
}
}
}
}
```

Output


12:40
Box Layout
Box Layout
Top End
Center
Bottom Start
Bottom End

Expt No : 6 **Date:**

App Navigation

Aim

Toimplement navigation between multiple screens in Android app.

Algorithm

1. Createmultiple activities/fragments. 2.
Use Intent to navigate.
3. Pass data using extras.
4. Implement Navigation component. 5.
Test navigation flow.

CODING

MainActivity.kt

```
package com.example.sai1
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button

classMainActivity : AppCompatActivity() {

   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)

      val btnPage1: Button = findViewById(R.id.btnPage1)
      val btnPage2: Button = findViewById(R.id.btnPage2)
      val btnPage3: Button = findViewById(R.id.btnPage3)
      val btnPage4: Button = findViewById(R.id.btnPage4)

      btnPage1.setOnClickListener {
         startActivity(Intent(this, Page1Activity::class.java))
      }
      btnPage2.setOnClickListener {

         startActivity(Intent(this, Page2Activity::class.java))
      }
      btnPage3.setOnClickListener {

         startActivity(Intent(this, Page3Activity::class.java))
      }
```

```
        btnPage4.setOnClickListener {
            startActivity(Intent(this, Page4Activity::class.java))
        }
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp"
    android:background="@color/white">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="GUI Components"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp" />

    <Button
        android:id="@+id/btnPage1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 1 - Initial State"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnPage2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 2 - Large Font"
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnPage3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 3 - Green Text"
```

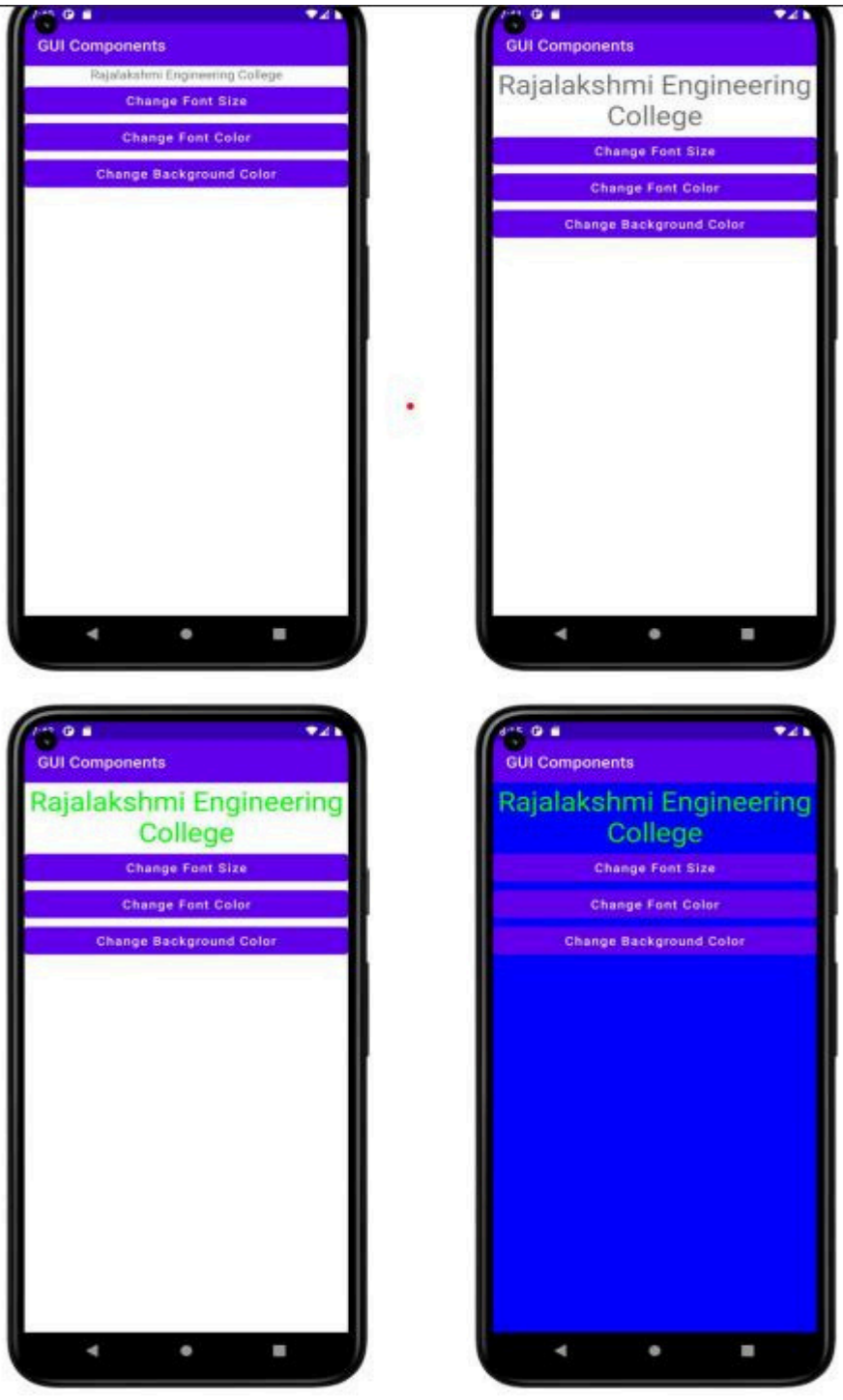
```
        android:textSize="16sp"
        android:layout_marginBottom="16dp" />

    <Button
        android:id="@+id/btnPage4"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Page 4 - Blue Background"
        android:textSize="16sp" />

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"
        android:theme="@style/Theme.Sai1"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".Page1Activity"
            android:exported="false" />
        <activity
            android:name=".Page2Activity"
            android:exported="false" />
        <activity
            android:name=".Page3Activity"
            android:exported="false" />
        <activity
            android:name=".Page4Activity"
            android:exported="false" />
    </application>
```

</manifest>

OUTPUT



Expt No : 7 **Date:**

Activity and Fragment Lifecycles

Aim

Tounderstand lifecycle methods of Activity and Fragment.

Algorithm

1. Createactivity/fragment.
2. Override lifecycle methods.
3. Use Logcat for logging.
4. Rotate device and observe lifecycle.
5. Analyze lifecycle behavior.

activity_main.xml

```
<?xmlversion="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context=".MainActivity">
<TextView
android:id="@+id/tvTitle"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Student Details"
android:textAlignment="center"
android:textSize="24sp" />
<LinearLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical">
<fragment
android:id="@+id/fragmentBasic"
android:name="org.rajalakshmi.androidfragments.StudentBasicDetails"
android:layout_width="match_parent"
android:layout_height="300dp" />
<fragment
android:id="@+id/fragmentMark"

android:name="org.rajalakshmi.androidfragments.StudentMarkDetails"
android:layout_width="match_parent"
android:layout_height="300dp" />
</LinearLayout>
```

```
</LinearLayout>
```

fragment_student_basic_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout    xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".StudentBasicDetails">
<TextView
android:id="@+id/tvBasicDetails"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Basic Details"
android:textAlignment="center"
android:textSize="24sp" />
<TextView
android:id="@+id/tvRegisterNumber"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="Register No." />
<EditText
android:id="@+id/etRegisterNumber"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:ems="10"
android:hint="Register Number"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvName"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="125dp"
android:text="Name" />
<EditText
android:id="@+id/etName"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="125dp"
android:ems="10"
android:hint="Name"
android:inputType="textPersonName" />
```

```
<TextView
android:id="@+id/tvDepartment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="200dp"
android:text="Department" />
<EditText
android:id="@+id/etDepartment"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="200dp"
android:ems="10"
android:hint="Department"
android:inputType="textPersonName"   />
</FrameLayout>
```

fragment_student_mark_details.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".StudentMarkDetails">
<TextView
android:id="@+id/tvBasicDetails"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Mark Details"
android:textAlignment="center"
android:textSize="24sp" />
<TextView
android:id="@+id/tvSSLC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="S.S.L.C." />
<EditText
android:id="@+id/etSSLC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:ems="10"
android:hint="S.S.L.C. Mark"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvHSc"
```

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="125dp"
android:text="H.Sc." />
<EditText
android:id="@+id/etHSC"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="125dp"
android:ems="10"
android:hint="H.Sc.                    Mark"
android:inputType="textPersonName"   />
<TextView
android:id="@+id/tvUG"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="200dp"
android:text="U.G." />
<EditText
android:id="@+id/etUG"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="200dp"
android:ems="10"
android:hint="U.G.                C.G.P.A."
android:inputType="textPersonName"   />
</FrameLayout>
```

MainActivity.kt

```
package org.rajalakshmi.androidfragments
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
}
}
```


StudentBasicDetails.kt

```
package org.rajalakshmi.androidfragments
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
```

```
import android.view.ViewGroup
// TODO: Rename parameter arguments, choose names that match
// the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private const val ARG_PARAM1 = "param1"
private const val ARG_PARAM2 = "param2"
/**
* A simple [Fragment] subclass.
* Use the [StudentBasicDetails.newInstance] factory method to
* create an instance of this fragment.
*/
class StudentBasicDetails : Fragment() {
// TODO: Rename and change types of parameters
private var param1: String? = null
private var param2: String? = null
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
arguments?.let {
param1 = it.getString(ARG_PARAM1)
param2 = it.getString(ARG_PARAM2)
}
}
override fun onCreateView(
inflater: LayoutInflater, container: ViewGroup?,
savedInstanceState: Bundle?
): View? {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_student_basic_details,
container, false)
}
companion object {
/**
* Use this factory method to create a new instance of
* this fragment using the provided parameters.
*
* @param param1 Parameter 1.
* @param param2 Parameter 2.
* @return A new instance of fragment StudentBasicDetails.
*/
// TODO: Rename and change types and number of parameters
@JvmStatic
fun newInstance(param1: String, param2: String) =
StudentBasicDetails().apply {
arguments = Bundle().apply {
putString(ARG_PARAM1, param1)
putString(ARG_PARAM2, param2)
}
}
}
```

```
}
```


StudentMarkDetails.kt

```
package org.rajalakshmi.androidfragments
import android.os.Bundle
import androidx.fragment.app.Fragment
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
// TODO: Rename parameter arguments, choose names that match
// the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private const val ARG_PARAM1 = "param1"
private const val ARG_PARAM2 = "param2"
/**
* A simple [Fragment] subclass.
* Use the [StudentMarkDetails.newInstance] factory method to
* create an instance of this fragment.
*/
class StudentMarkDetails : Fragment() {
// TODO: Rename and change types of parameters
private var param1: String? = null
private var param2: String? = null
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
arguments?.let {
param1 = it.getString(ARG_PARAM1)
param2 = it.getString(ARG_PARAM2)
}
}
override fun onCreateView(
inflater: LayoutInflater, container: ViewGroup?,
savedInstanceState: Bundle?
): View? {
// Inflate the layout for this fragment
return inflater.inflate(R.layout.fragment_student_mark_details,
container, false)
}
companion object {
/**
* Use this factory method to create a new instance of
* this fragment using the provided parameters.
*
* @param param1 Parameter 1.
* @param param2 Parameter 2.
* @return A new instance of fragment StudentMarkDetails.
*/
// TODO: Rename and change types and number of parameters
```

```
@JvmStatic
fun newInstance(param1: String, param2: String) =
StudentMarkDetails().apply {
arguments = Bundle().apply {
putString(ARG_PARAM1, param1)
putString(ARG_PARAM2, param2)
}
}
}
}
```

Output

Output

Expt No : 8 **Date:**

App Architecture (UI Layer)

Aim

Toimplement UI layer using MVVM architecture.

Algorithm

1. Create ViewModel class.
2. Add LiveData objects.
3. Bind UI with ViewModel.
4. Observe data changes.
5. Update UI automatically.

Coding

Mainactivity.kt

```
packagecom.example.apparchitectureuilayer

importandroid.os.Bundle
importandroidx.activity.ComponentActivity
importandroidx.activity.compose.setContent
importandroidx.activity.enableEdgeToEdge
importcom.example.apparchitectureuilayer.presentation.navigation.AppNavigationSimple
importcom.example.apparchitectureuilayer.ui.theme.AppArchitectureUILayerTheme

classMainActivity : ComponentActivity() {


    overridefun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            AppArchitectureUILayerTheme {
                AppNavigationSimple()
            }
        }
    }
}
```

UserDetailScreen.kt

```
packagecom.example.apparchitectureuilayer.presentation.screen

importandroidx.compose.foundation.background
importandroidx.compose.foundation.layout.*
importandroidx.compose.foundation.lazy.LazyColumn
importandroidx.compose.foundation.shape.CircleShape
importandroidx.compose.material3.*
```

```
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.layout.ContentScale
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.hilt.navigation.compose.hiltViewModel
import coil.compose.AsyncImage
import com.example.apparchitectureuilayer.data.model.ApiResponse
import
com.example.apparchitectureuilayer.presentation.viewmodel.UserDetailViewModel

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun UserDetailScreen(
    userId: String,
    onBack: () -> Unit,
    viewModel: UserDetailViewModel = hiltViewModel()
) {
    LaunchedEffect(userId) {
        viewModel.loadUser(userId)
    }

    val user by viewModel.user.collectAsState()

    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("User Details") },
                navigationIcon = {
                    IconButton(onClick = onBack) {
                        Text("←")
                    }
                }
            )
        }
    ) { padding ->
        Box(
            modifier = Modifier
                .fillMaxSize()
                .padding(padding)
        ) {
            when (user) {
                is ApiResponse.Loading -> {
                    CircularProgressIndicator(
                        modifier = Modifier.align(Alignment.Center)
                    )
```

```
        }
        is ApiResponse.Success -> {
            valuserData=(userasApiResponse.Success).data
            LazyColumn(
                modifier=Modifier.fillMaxSize(),
                contentPadding=PaddingValues(16.dp),
                verticalArrangement=Arrangement.spacedBy(16.dp)
            ) {
                item{
                    Row(
                        modifier=Modifier.fillMaxWidth(),
                        horizontalArrangement=Arrangement.Center
                    ) {
                        if(userData.avatar!=null){
                            AsyncImage(
                                model=userData.avatar,
                                contentDescription="User Avatar",
                                modifier = Modifier
                                    .size(120.dp)
                                    .clip(CircleShape ),
                                contentScale=ContentScale.Crop
                            )
                        } else {
                            Box(
                                modifier = Modifier
                                    .size(120.dp)
                                    .clip(CircleShape )
                                    .background(MaterialTheme.colorScheme.primaryContainer),
                                contentAlignment=Alignment.Center
                            ) {
                                Text(
                                    text=userData.name.first().uppercase(),
                                    style=MaterialTheme.typography.headlineLarge,
                                    color=MaterialTheme.colorScheme.onPrimaryContainer
                                )
                            }
                        }
                    }
                }

                item{
                    Column(
                        horizontalAlignment=Alignment.CenterHorizontally
                    ) {
                        Text(
                            text = userData.name,
                            style=MaterialTheme.typography.headlineMedium,
                            fontWeight=FontWeight.Bold
```

```
            )
            Spacer(modifier=Modifier.height(8.dp))
            Text(
                text=userData.email,
                style=MaterialTheme.typography.bodyLarge,
                color=MaterialTheme.colorScheme.onSurfaceVariant
            )
            Spacer(modifier=Modifier.height(8.dp))
            Text(
                text=if(userData.isActive) "Active" else "Inactive",
                style=MaterialTheme.typography.bodyMedium,
                color=if(userData.isActive)
                    MaterialTheme.colorScheme.primary else
                    MaterialTheme.colorScheme.error,
                modifier=Modifier
                    .padding(horizontal = 16.dp)
                    .padding(vertical = 4.dp)
                    .background(
                        color=if(userData.isActive)
                            MaterialTheme.colorScheme.primaryContainer else
                            MaterialTheme.colorScheme.errorContainer,
                        shape=MaterialTheme.shapes.small
                    )
                    .padding(horizontal = 12.dp, vertical = 4.dp)
            )
        }
    }

    item{
        Card(
            modifier=Modifier.fillMaxWidth()
        ) {
            Column(
                modifier=Modifier
                .fillMaxWidth()
                .padding(16.dp)
            ) {
                Text(
                    text="UserInformation",
                    style=MaterialTheme.typography.titleMedium,
                    fontWeight=FontWeight.Bold
                )
                Spacer(modifier= Modifier.height(12.dp))

                InfoRow("ID",userData.id)
                InfoRow("Name", userData.name)
                InfoRow("Email", userData.email)
                InfoRow("Status", if (userData.isActive) "Active" else "Inactive")
```

```
                    }
                }
            }
        }
    }
    isApiResponse.Error -> {
        Column(
            modifier=Modifier.align(Alignment.Center),
            horizontalAlignment = Alignment.CenterHorizontally
        ) {
            Text(
                text="Error: ${(user as ApiResponse.Error).message}",
                color=MaterialTheme.colorScheme.error
            )
            Spacer(modifier = Modifier.height(8.dp))
            Button(onClick = { viewModel.loadUser(userId) }) {
                Text("Retry")
            }
        }
    }
}
}
}

@Composable
privatefunInfoRow(label:String, value: String) {
    Column {
        Text(
            text =label,
            style=MaterialTheme.typography.bodySmall,
            color=MaterialTheme.colorScheme.onSurfaceVariant
        )
        Text(
            text =value,
            style=MaterialTheme.typography.bodyMedium,
            modifier=Modifier.padding(top = 2.dp)
        )
        Spacer(modifier=Modifier.height(8.dp))
    }
}
```

AppNavigationSimple.kt

```
packagecom.example.apparchitectureuilayer.presentation.navigation
importandroidx.compose.foundation.background
importandroidx.compose.foundation.layout.*
importandroidx.compose.foundation.shape.RoundedCornerShape
```

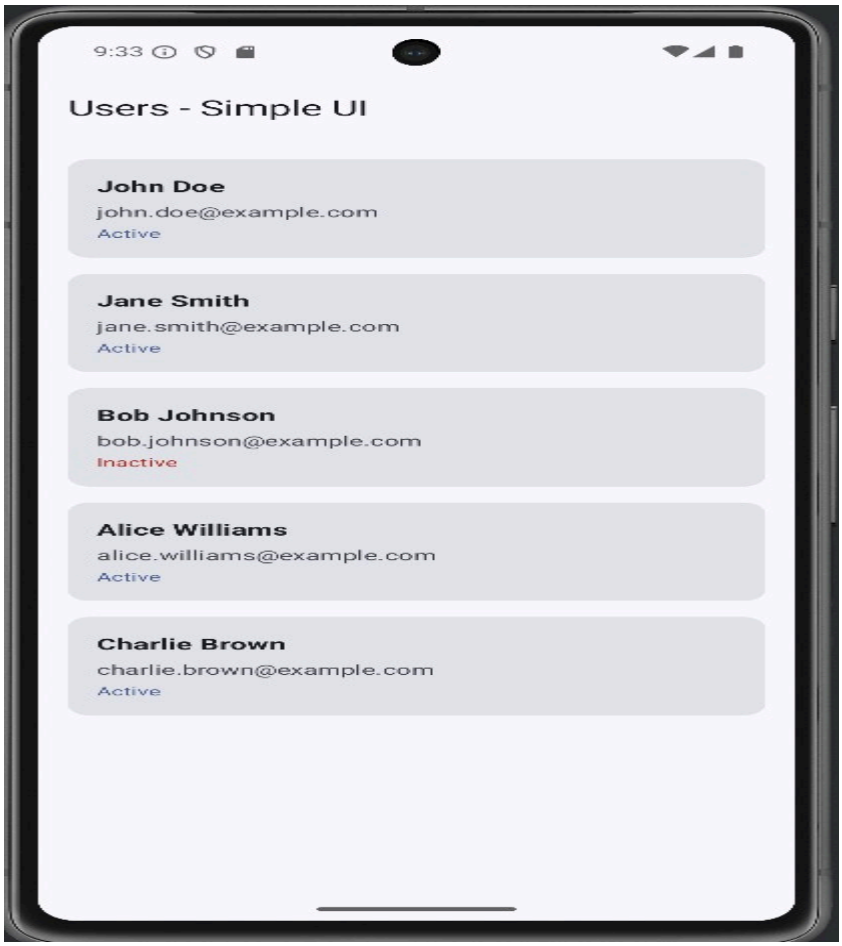
```
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Alignment
import androidx.compose.ui.Modifier
import androidx.compose.ui.draw.clip
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import com.example.apparchitectureuilayer.data.model.User
import com.example.apparchitectureuilayer.presentation.screen.UserListScreenSimple

@Composable
fun AppNavigationSimple() {
    var currentUser by remember { mutableStateOf<User?>(null) }

    if (currentUser == null) {
        UserListScreenSimple(
            onUserClick = { userId ->
                // Find user and show details
                val users = listOf(
                    User(
                        id = "1",
                        name = "John Doe",
                        email = "john.doe@example.com",
                        avatar = null,
                        isActive = true
                    ),
                    User(
                        id = "2",
                        name = "Jane Smith",
                        email = "jane.smith@example.com",
                        avatar = null,
                        isActive = true
                    ),
                    User(
                        id = "3",
                        name = "Bob Johnson",
                        email = "bob.johnson@example.com",
                        avatar = null,
                        isActive = false
                    ),
                    User(
                        id = "4",
                        name = "Alice Williams",
                        email = "alice.williams@example.com",
                        avatar = null,
                        isActive = true
                    ),
                    User(
```

```
                    id = "5",
                    name = "Charlie Brown",
                    email = "charlie.brown@example.com",
                    avatar = null,
                    isActive = true
                )
            )
            currentUser = users.find { it.id == userId }
        }
    )
} else {
    UserDetailScreenSimple(
    user = currentUser!!,
    onBack = { currentUser = null }
    )
}
}

@OptIn(ExperimentalMaterial3Api::class)
@Composable
fun UserDetailScreenSimple(
    user: User,
    onBack: () -> Unit
) {
    Scaffold(
        topBar = {
            TopAppBar(
                title = { Text("User Details") },
                navigationIcon = {
                    Button(onClick = onBack) {
                        Text("← Back")
                    }
                }
            )
        }
    ) { padding ->
        Column(
            modifier = Modifier
                .fillMaxSize()
                .padding(padding)
                .padding(16.dp),
            verticalArrangement = Arrangement.spacedBy(16.dp)
        ) {
            // User avatar placeholder
            Box(
                modifier = Modifier
                    .size(120.dp)
                    .align(Alignment.CenterHorizontally)
```

```
                .clip(RoundedCornerShape(16.dp))
                .background(
                    color=MaterialTheme.colorScheme.primaryContainer
                ),
            contentAlignment = Alignment.Center
        ) {
            Text(
                text=user.name.first().uppercase(),
                style=MaterialTheme.typography.headlineLarge,
                color=MaterialTheme.colorScheme.onPrimaryContainer
            )
        }

        // User info
        Card(
            modifier=Modifier.fillMaxWidth()
        ) {
            Column(
                modifier=Modifier
                .fillMaxWidth()
                .padding(16.dp),
                verticalArrangement = Arrangement.spacedBy(8.dp)
            ) {
                Text(
                    text=user.name,
                    style=MaterialTheme.typography.headlineMedium,
                    fontWeight= FontWeight.Bold
                )
                Text(
                    text=user.email,
                    style=MaterialTheme.typography.bodyLarge,
                    color=MaterialTheme.colorScheme.onSurfaceVariant
                )
                Text(
                    text="Status: ${if (user.isActive) "Active" else "Inactive"}",
                    style=MaterialTheme.typography.bodyMedium,
                    color=if(user.isActive)
                        MaterialTheme.colorScheme.primary else
                        MaterialTheme.colorScheme.error
                )
                Text(
                    text="ID:${user.id}",
                    style=MaterialTheme.typography.bodySmall,
                    color=MaterialTheme.colorScheme.onSurfaceVariant
                )
            }
        }
    }
```

```
    }
}
```

Output


9:33
Users - Simple UI
John Doe
john.doe@example.com
Active
Jane Smith
jane.smith@example.com
Active
Bob Johnson
bob.johnson@example.com
Inactive
Alice Williams
alice.williams@example.com
Active
Charlie Brown
charlie.brown@example.com
Active


9:34
← Back
User Details
J
John Doe
john.doe@example.com
Status: Active
ID: 1

Expt No : 9 **Date:**

App Architecture (Persistence)

Aim

Tostore and retrieve data using Room persistence library.

Algorithm

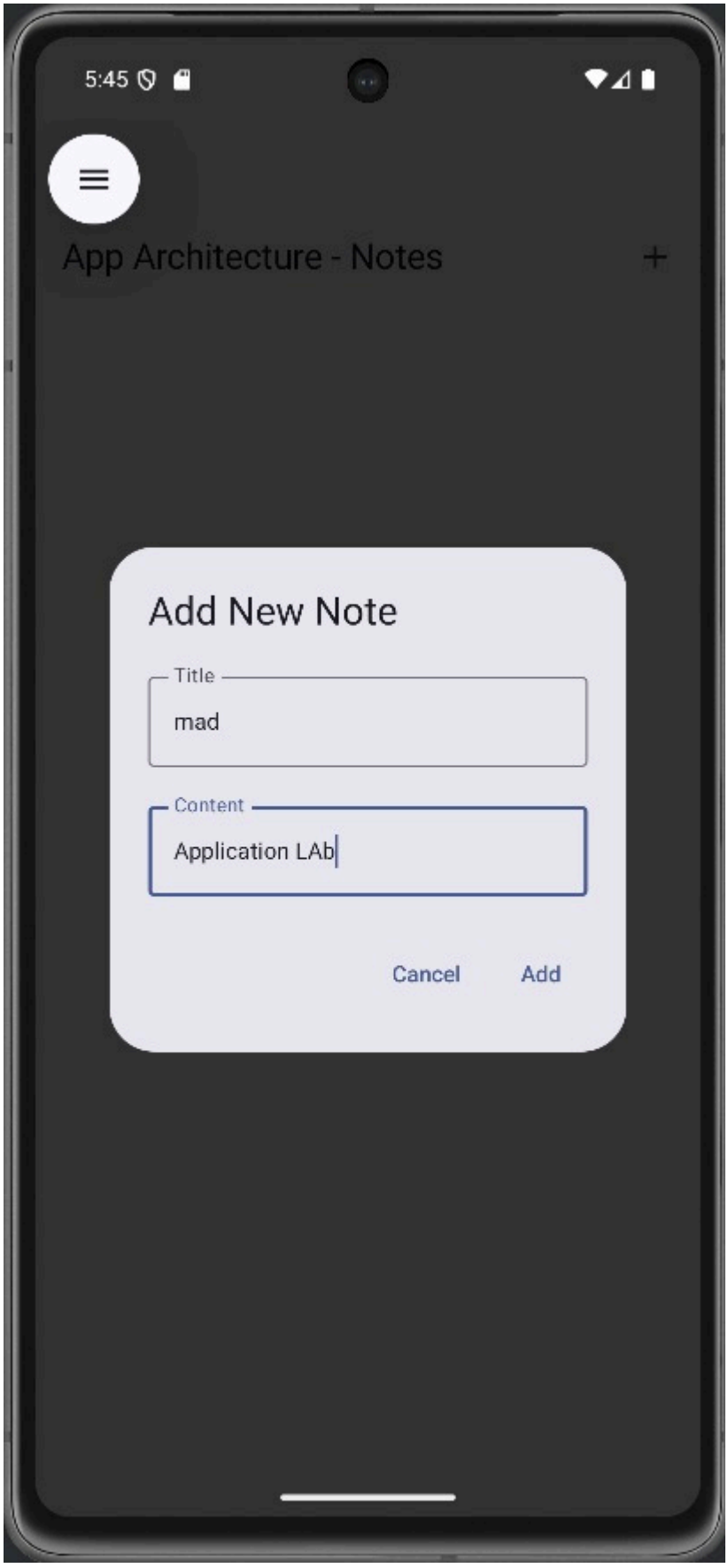
1. AddRoom dependencies. 2.
Create Entity class.
3. Create DAO interface.
4. Create Database class.
5. Perform CRUD operations.

MainActivity.kt

```
package com.example.apparchitecturepersistence
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.foundation.layout.padding
import androidx.compose.material3.Scaffold
import androidx.compose.runtime.Composable
import androidx.compose.runtime.collectAsState
import androidx.compose.runtime.getValue
import androidx.compose.ui.Modifier
import androidx.compose.ui.tooling.preview.Preview
import androidx.lifecycle.viewmodel.compose.viewModel
import androidx.compose.runtime.livedata.observeAsState
import com.example.apparchitecturepersistence.data.NoteDatabase
import com.example.apparchitecturepersistence.data.Note
import com.example.apparchitecturepersistence.repository.NoteRepository
import
com.example.apparchitecturepersistence.ui.theme.AppArchitecturePersistenceTheme
import com.example.apparchitecturepersistence.ui.screens.NoteListScreen
import com.example.apparchitecturepersistence.viewmodel.NoteViewModel
import com.example.apparchitecturepersistence.viewmodel.NoteViewModelFactory

classMainActivity : ComponentActivity() {

   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      enableEdgeToEdge()

      val database = NoteDatabase.getDatabase(this)
      val repository = NoteRepository(database.noteDao())
```

```
        setContent {
            AppArchitecturePersistenceTheme {
                Scaffold(modifier = Modifier.fillMaxSize()) { innerPadding ->
                    NoteApp(
                        repository = repository,
                        modifier = Modifier.padding(innerPadding)
                    )
                }
            }
        }
    }
}

@Composable
fun NoteApp(repository: NoteRepository, modifier: Modifier = Modifier) {
    val viewModel: NoteViewModel = viewModel(
        factory = NoteViewModelFactory(repository)
    )

    val notes by viewModel.allNotes.observeAsState(initial = emptyList())
    val selectedNote by viewModel.selectedNote.observeAsState()
    val isLoading by viewModel.isLoading.observeAsState(initial = false)
    val errorMessage by viewModel.errorMessage.observeAsState()

    NoteListScreen(
        notes = notes ?: emptyList(),
        selectedNote = selectedNote,
        isLoading = isLoading ?: false,
        errorMessage = errorMessage,
        onNoteClick = { note -> viewModel.selectNote(note) },
        onNoteDelete = { note -> viewModel.deleteNote(note) },
        onNoteToggle = { note -> viewModel.toggleNoteCompletion(note) },
        onAddNote = { title, content -> viewModel.insertNote(title, content) },
        onUpdateNote = { note -> viewModel.updateNote(note) },
        onClearError = { viewModel.clearError() },
        onDeleteAll = { viewModel.deleteAllNotes() },
        formatTimestamp = { timestamp -> viewModel.formatTimestamp(timestamp) },
        modifier = modifier
    )
}

@Preview(showBackground = true)
@Composable
funNoteAppPreview() {
    AppArchitecturePersistenceTheme {
        // Preview would need mock repository for full functionality
    }
}
```

Output


5:45
App Architecture - Notes
Add New Note
Title
mad
Content
Application LAb
Cancel
Add

Expt No : 10 **Date:**

Advanced RecyclerView Use Cases

Aim
Todisplay complex and dynamic lists using RecyclerView.

Algorithm
1. AddRecyclerView to layout.
2. Create ViewHolder.
3. Create Adapter.
4. Bind data list.
5. Handle item clicks.

```
package com.example.advancedrecyclerviewusecases

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.ItemTouchHelper
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView
import
com.example.advancedrecyclerviewusecases.adapter.AdvancedRecyclerViewAdapter
import
com.example.advancedrecyclerviewusecases.helper.ItemTouchHelperCallback
import com.example.advancedrecyclerviewusecases.model.DataProvider
import com.example.advancedrecyclerviewusecases.model.Item
import com.google.android.material.snackbar.Snackbar

class MainActivity : AppCompatActivity() {

    private lateinit var recyclerView: RecyclerView
    private lateinit var adapter: AdvancedRecyclerViewAdapter
    private lateinit var itemTouchHelper: ItemTouchHelper
    private var items = DataProvider.getSampleData()

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        setupRecyclerView()
        setupSwipeToDelete()
        setupDragAndDrop()
    }
```

```
privatefun setupRecyclerView() {
    recyclerView = findViewById(R.id.recyclerView)
    adapter = AdvancedRecyclerViewAdapter(
        items = items,
        onItemClick = { item ->
            handleItemClick(item)
        },
        onDeleteItem = { item ->
            handleDeleteItem(item)
        }
    )

    recyclerView.apply {
        layoutManager = LinearLayoutManager(this@MainActivity)
        adapter = this@MainActivity.adapter
    }
}

privatefun handleItemClick(item: Item) {
    Snackbar.make(
        recyclerView,
        "Clicked: ${item.title}",
        Snackbar.LENGTH_SHORT
    ).show()
}

privatefun handleDeleteItem(item: Item) {
    valposition = items.indexOf(item)
    if(position != -1) {
        items = items.toMutableList().apply { removeAt(position) }
        adapter.removeItem(item)

        Snackbar.make(
            recyclerView,
            "Deleted: ${item.title}",
            Snackbar.LENGTH_LONG
        ).setAction("Undo") {
            items = items.toMutableList().apply { add(position, item) }
            adapter.updateItems(items)
        }.show()
    }
}

privatefun setupSwipeToDelete() {
    valcallback = ItemTouchHelperCallback(
        onItemMove = { fromPosition, toPosition ->
            //Handle item move for drag and drop
```

```
            valmovedItem = items[fromPosition]
            items=items.toMutableList().apply {
                removeAt(fromPosition)
                add(toPosition, movedItem)
            }
            adapter.notifyItemMoved(fromPosition, toPosition)
        },
        onItemSwiped = { position ->
            //Handle item swipe for delete
            valswipedItem = items[position]
            handleDeleteItem(swipedItem)
        }
    )

    itemTouchHelper = ItemTouchHelper(callback)
    itemTouchHelper.attachToRecyclerView(recyclerView)
  }

  privatefunsetupDragAndDrop() {
    //Draganddrop is already set up in setupSwipeToDelete()
    //TheItemTouchHelper handles both swipe and drag operations
  }
}
```

Output

Expt No :1 1 **Date:**

Connect to the Internet

Aim

Tofetch data from internet using REST API.

Algorithm

1. Add internet permission.
2. Add Retrofit dependency.
3. Create API interface.
4. Make network call.
5. Display response data.

Coding

AndroidManifest.xml

```
<?xmlversion="1.0"encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
<application
android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:supportsRtl="true"
android:theme="@style/Theme.SendEmail"
tools:targetApi="31">
<activity
android:name=".MainActivity"
android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

activity_main.xml

```
<?xmlversion="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
```

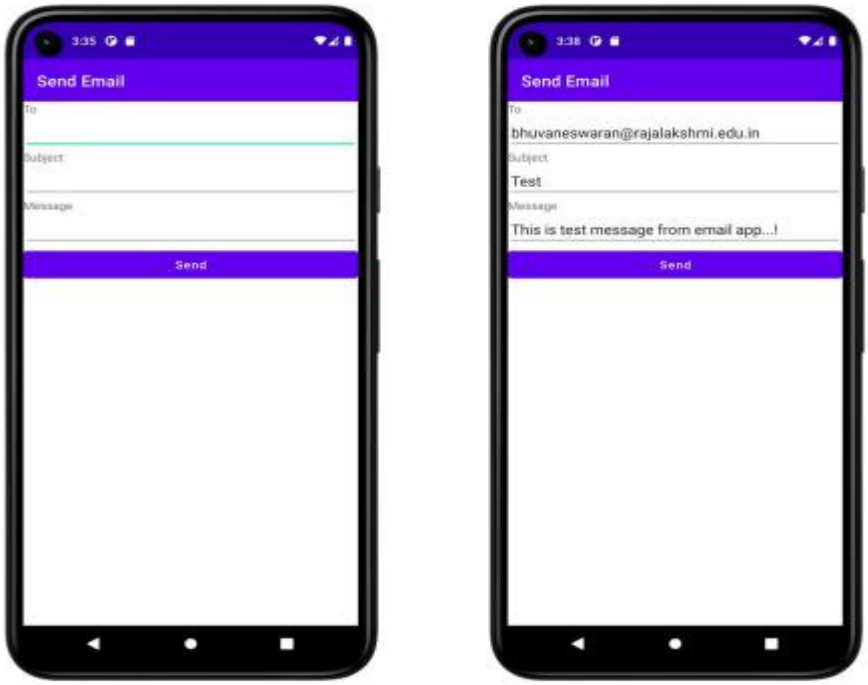
```
tools:context=".MainActivity">
<TextView
android:id="@+id/tvEmail"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="To" />
<EditText
android:id="@+id/etEmail"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvSubject"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Subject" />
<EditText
android:id="@+id/etSubject"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<TextView
android:id="@+id/tvMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Message" />
<EditText
android:id="@+id/etMessage"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:ems="10"
android:inputType="textPersonName" />
<Button
android:id="@+id/btSend"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Send"
android:textAllCaps="false" />
</LinearLayout>
```

MainActivity.kt

```
package org.rajalakshmi.sendemail
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
```

```
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
class MainActivity : AppCompatActivity() {
override fun onCreate(savedInstanceState: Bundle?) {
super.onCreate(savedInstanceState)
setContentView(R.layout.activity_main)
val etEmail : EditText = findViewById(R.id.etEmail)
val etSubject : EditText = findViewById(R.id.etSubject)
val etMessage : EditText = findViewById(R.id.etMessage)
val btSend : Button = findViewById(R.id.btSend)
btSend.setOnClickListener {
val email = etEmail.text.toString()
val subject = etSubject.text.toString()
val message = etMessage.text.toString()
val intent = Intent(Intent.ACTION_SEND)
intent.putExtra(Intent.EXTRA_EMAIL, arrayOf(email))
intent.putExtra(Intent.EXTRA_SUBJECT, subject)
intent.putExtra(Intent.EXTRA_TEXT, message)
intent.type = "message/rfc822"
startActivity(Intent.createChooser(intent, "Choose an Email client :"))
}
}
}
```

Output



Expt No : 12 **Date:**

Repository Patternand WorkManager

Aim

Tomanage data sources and background tasks efficiently.

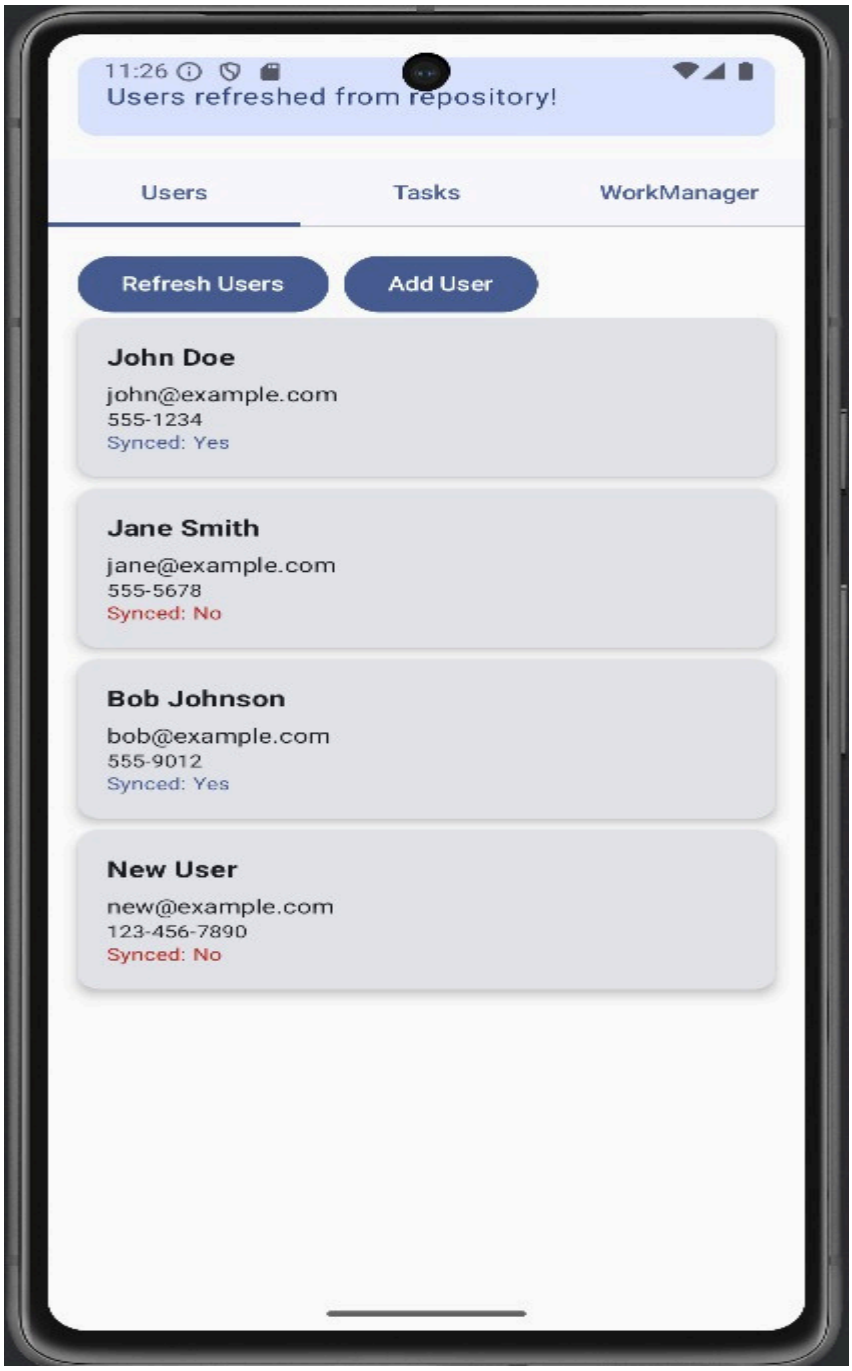
Algorithm

1. Createrepository class.
2. Connect repository to ViewModel.
3. Add WorkManager dependency.
4. Create Worker class.
5. Schedule background task.

Coding

```
package com.example.repository

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.activity.enableEdgeToEdge
import com.example.repository.ui.theme.RepositoryTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContent {
            RepositoryTheme {
                RepositoryDemoScreen()
            }
        }
    }
}
```

Output


11:26
Users refreshed from repository!
Users
Tasks
WorkManager
Refresh Users
Add User
John Doe
john@example.com
555-1234
Synced: Yes
Jane Smith
jane@example.com
555-5678
Synced: No
Bob Johnson
bob@example.com
555-9012
Synced: Yes
New User
new@example.com
123-456-7890
Synced: No

Expt No : 13 **Date:**

App UIDesign

Aim

Todesign attractive, responsive, and user-friendly Android UI.

Algorithm

1. ApplyMaterial Design principles.
2. Use themes and styles.
3. Add colors and fonts.
4. Use icons and animations.
5. Test UI responsiveness.

MainActivity.kt

```
package com.example.japp9

import android.os.Bundle
import android.os.Handler
import android.os.Looper
import android.text.Editable
import android.text.TextWatcher
import android.widget.Button
import android.widget.EditText
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView
import com.example.japp9.adapter.AdvancedRecyclerViewAdapter
import com.example.japp9.model.ExpandableItemModel
import com.example.japp9.model.RecyclerViewItem
import com.example.japp9.model.User
import com.example.japp9.utils.DragAndDropCallback
import com.example.japp9.utils.FilterUtils
import com.example.japp9.utils.InfiniteScrollListener
import com.example.japp9.utils.SwipeToDeleteCallback

class MainActivity : AppCompatActivity() {


    private lateinit var recyclerView: RecyclerView
    private lateinit var adapter: AdvancedRecyclerViewAdapter
    private lateinit var searchEditText: EditText
    private lateinit var toggleSwipeButton: Button
```

```
    private lateinit var toggleDragButton: Button

    private var allItems = mutableListOf<RecyclerViewItem>()
    private var currentItems = mutableListOf<RecyclerViewItem>()
    private var swipeToDeleteHelper: androidx.recyclerview.widget.ItemTouchHelper? =
null
    private var dragAndDropHelper: androidx.recyclerview.widget.ItemTouchHelper? =
null
    private var infiniteScrollListener: InfiniteScrollListener? = null

    private var isLoading = false
    private var currentPage = 1
    private val itemsPerPage = 10

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        initViews()
        setupRecyclerView()
        generateInitialData()
        setupListeners()
        setupGestures()
    }

    private fun initViews() {
        recyclerView = findViewById(R.id.recyclerView)
        searchEditText = findViewById(R.id.searchEditText)
        toggleSwipeButton = findViewById(R.id.toggleSwipeButton)
        toggleDragButton = findViewById(R.id.toggleDragButton)
    }

    private fun setupRecyclerView() {
        adapter = AdvancedRecyclerViewAdapter(
            items = currentItems,
            onItemClick = { item ->
                handleItemClick(item)
            },
            onDeleteItem = { position ->
                adapter.removeItem(position)
            }
        )

        recyclerView.layoutManager = LinearLayoutManager(this)
        recyclerView.adapter = adapter

        // Setup infinite scrolling
        val layoutManager = recyclerView.layoutManager as LinearLayoutManager
```

```
        infiniteScrollListener = object : InfiniteScrollListener(layoutManager) {
            override fun onLoadMore() {
                loadMoreItems()
            }
        }
        recyclerView.addOnScrollListener(infiniteScrollListener!!)
    }

    privatefun generateInitialData() {
        //Generate sample users
        valusers = (1..20).map { index ->
            User(
                id = index,
                name = "User $index",
                email = "user$index@example.com",
                avatar = "",
                isActive = index % 2 == 0
            )
        }
        //Generate sample expandable items
        valexpandableItems = (1..5).map { index ->

            ExpandableItemModel(
                id = index + 100,
                title = "Category $index",
                description = "Description for category $index",
                subItems = (1..3).map { subIndex -> "Sub item $subIndex in category $index" }
            )
        }
        allItems.clear()
        allItems.add(RecyclerViewItem.HeaderItem("Active Users"))
        users.filter { it.isActive }.take(5).forEach {

            allItems.add(RecyclerViewItem.UserItem(it))
        }
        allItems.add(RecyclerViewItem.HeaderItem("Categories"))
        expandableItems.forEach {

            allItems.add(RecyclerViewItem.ExpandableItem(it))
        }

        allItems.add(RecyclerViewItem.HeaderItem("All Users"))
        users.forEach {
            allItems.add(RecyclerViewItem.UserItem(it))
        }

        currentItems.clear()
        currentItems.addAll(allItems.take(itemsPerPage))
```

```
        adapter.updateItems(currentItems)
    }

    privatefun setupListeners() {
        searchEditText.addTextChangedListener(object : TextWatcher {
            override fun beforeTextChanged(s: CharSequence?, start: Int, count: Int, after: Int)
{}
            override fun onTextChanged(s: CharSequence?, start: Int, before: Int, count: Int)
{}
            override fun afterTextChanged(s: Editable?) {
                filterItems(s?.toString() ?: "")
            }
        })

        toggleSwipeButton.setOnClickListener {
            toggleSwipeToDelete()
        }
        toggleDragButton.setOnClickListener {

            toggleDragAndDrop()
        }
    }

    private fun setupGestures() {
        // Setup swipe to delete
        val swipeCallback = SwipeToDeleteCallback { position ->
            adapter.removeItem(position)
        }
        swipeToDeleteHelper =
androidx.recyclerview.widget.ItemTouchHelper(swipeCallback)
        swipeToDeleteHelper?.attachToRecyclerView(recyclerView)

        // Setup drag and drop
        val dragCallback = DragAndDropCallback { fromPosition, toPosition ->
            adapter.moveItem(fromPosition, toPosition)
        }
        dragAndDropHelper =
androidx.recyclerview.widget.ItemTouchHelper(dragCallback)
        // Don't attach by default, let user toggle it
    }

    private fun handleItemClick(item: RecyclerViewItem) {
        when (item) {
            is RecyclerViewItem.ExpandableItem -> {
                // Toggle expandable item
                val updatedItem = item.item.copy(isExpanded = !item.item.isExpanded)
                val index = currentItems.indexOf(item)
                if (index != -1) {
```

```
                currentItems[index] = RecyclerViewItem.ExpandableItem(updatedItem)
                adapter.notifyItemChanged(index)
            }
        }
        is RecyclerViewItem.UserItem -> {
            // Handle user click - could show details or edit
            println("Clicked on user: ${item.user.name}")
        }
        else -> {}
    }
}

private fun filterItems(query: String) {
    val filtered = FilterUtils.filterItems(allItems, query)
    currentItems.clear()
    currentItems.addAll(filtered.take(itemsPerPage))
    adapter.updateItems(currentItems)
}

private fun toggleSwipeToDelete() {
    if (swipeToDeleteHelper == null) {
        val swipeCallback = SwipeToDeleteCallback { position ->
            adapter.removeItem(position)
        }
        swipeToDeleteHelper =
androidx.recyclerview.widget.ItemTouchHelper(swipeCallback)
        swipeToDeleteHelper?.attachToRecyclerView(recyclerView)
        toggleSwipeButton.text = "Disable Swipe"
    } else {
        swipeToDeleteHelper?.attachToRecyclerView(null)
        swipeToDeleteHelper = null
        toggleSwipeButton.text = "Enable Swipe"
    }
}

private fun toggleDragAndDrop() {
    if (dragAndDropHelper == null) {
        val dragCallback = DragAndDropCallback { fromPosition, toPosition ->
            adapter.moveItem(fromPosition, toPosition)
        }
        dragAndDropHelper =
androidx.recyclerview.widget.ItemTouchHelper(dragCallback)
        dragAndDropHelper?.attachToRecyclerView(recyclerView)
        toggleDragButton.text = "Disable Drag"
    } else {
        dragAndDropHelper?.attachToRecyclerView(null)
        dragAndDropHelper = null
        toggleDragButton.text = "Enable Drag"
```

```
        }
    }

    privatefun loadMoreItems() {
        if(isLoading) return

        isLoading = true

        //Addloading indicator
        valloadingList = currentItems.toMutableList()
        loadingList.add(RecyclerViewItem.LoadingItem())
        adapter.updateItems(loadingList)

        //Simulate network delay
        Handler(Looper.getMainLooper()).postDelayed({
        valstartIndex = currentPage * itemsPerPage
        valendIndex = (startIndex + itemsPerPage).coerceAtMost(allItems.size)

        if(startIndex < allItems.size) {
            currentItems.addAll(allItems.subList(startIndex, endIndex))
            currentPage++

            //Remove loading indicator and add new items
            adapter.updateItems(currentItems)
            infiniteScrollListener?.setLoaded()
          }

          isLoading = false
        },1500)
    }
}
```


10:05
App UI Designer
Components
Button
Text Field
Text
Image
Card
Container
Properties
Select an element to edit properties
Elements


10:07
App UI Designer
Components
Button
Text Field
Text
Image
Card
Container
Properties
Text
Button
Width
100
Height
50
Corner Radius
8
Red
Blue
Green
Elements
BUTTON (element_1)